

Table of Contents

Uralstech.UXR.QuestCamera	3
CameraDevice	6
CameraDevice.ErrorCode	16
CameraDevice.Proxy	18
CameraInfo	20
CameraInfo.CameraEye	28
CameraInfo.CameraIntrinsics	29
CameraInfo.CameraSource	32
CameraMetadata	33
CameraMetadata.FloatRange	38
CameraMetadata.IntRange	40
CameraMetadata.Key	42
CameraMetadata.LongRange	47
CaptureFailure	49
CaptureFailure.ErrorCode	52
CapturePipeline<T>	53
CaptureRequest	55
CaptureRequest.Builder	57
CaptureResult	60
CaptureSessionBase<TProxy>	63
CaptureSessionBase<TProxy>.ErrorCode	69
CaptureSessionBase<TProxy>.ModifyRequestBuilderCallback	70
CaptureSessionBase<TProxy>.ProxyBase	71
CaptureSessionBase<TProxy>.ShouldRegisterCaptureEventsCallback	77
CaptureTemplate	78
ComputeShaderKernel	79
ContinuousCaptureSession	82
ContinuousCaptureSession.OnFrameReadyCallback	84
ContinuousCaptureSession.Proxy	86
JNIExtensions	88
OnDemandCaptureSession	98
OnDemandCaptureSession.RequestStatus	101
PCASupport	103
QuestCameraManager	104
ResourceState	110
StatefulResource	111
StreamUseCase	114
TotalCaptureResult	115

YUVConverter	117
Uralstech.UXR.QuestCamera.GLES	122
GLESAPI	123
GLESCaptureSession	127
GLESCaptureSession.Proxy	132
RenderJobDisposeData	133
RenderJobDisposeData.Callback	135
RenderJobEvent	136
RenderJobRunData	137
RenderJobRunData.Callback	139
RenderJobSetupData	140
RenderJobSetupData.Callback	142
Uralstech.UXR.QuestCamera.Vulkan	143
ManagedRenderCallback	144
RenderData	146
RenderData.Callback	149
VulkanAPI	150
VulkanContinuousCaptureSession	153
VulkanContinuousCaptureSession.OnFrameReadyCallback	158
VulkanContinuousCaptureSession.Proxy	159
VulkanOnDemandCaptureSession	161

Namespace Uralstech.UXR.QuestCamera

Classes

[CameraDevice](#)

Manages a camera device.

[CameraDevice.Proxy](#)

Java proxy to handle native callbacks.

[CameraInfo](#)

Information describing a camera device.

[CameraInfo.CameraIntrinsics](#)

Defines the camera's intrinsic properties. All values are in pixels.

[CameraMetadata](#)

Represents a map of camera device, capture request or capture result metadata.

[CameraMetadata.FloatRange](#)

A range of two float values.

[CameraMetadata.IntRange](#)

A range of two integer values.

[CameraMetadata.Key](#)

Light wrapper for [CaptureRequest.Key](#)[↗], [CameraCharacteristics.Key](#)[↗], etc.

[CameraMetadata.LongRange](#)

A range of two long values.

[CaptureFailure](#)

A report of failed capture for a single image capture from the image sensor.

[CapturePipeline<T>](#)

A wrapper for a complete capture-conversion pipeline.

[CaptureRequest](#)

An immutable package of settings and outputs needed to capture a single image from the camera device.

[CaptureRequest.Builder](#)

Builder for a capture request.

[CaptureResult](#)

The subset of the results of a single image capture from the image sensor.

[CaptureSessionBase<TProxy>](#)

Base class for all capture session types.

[CaptureSessionBase<TProxy>.ProxyBase](#)

Java proxy to handle native callbacks.

[ComputeShaderKernel](#)

Utility class to contain ComputeShader kernel info.

[ContinuousCaptureSession](#)

Manages a camera capture session with a continuous/repeating request.

[ContinuousCaptureSession.Proxy](#)

Java proxy to handle native callbacks.

[JNIExtensions](#)

QOL extensions for the JNI.

[OnDemandCaptureSession](#)

Manages an on-demand, explicit capture session.

[QuestCameraManager](#)

Entry point for the native Camera2 plugin.

[StatefulResource](#)

[TotalCaptureResult](#)

The total assembled results of a single image capture from the image sensor.

[YUVConverter](#)

Converts raw camera capture session frames to Unity-supported RGBA.

Structs

[OnDemandCaptureSession.RequestStatus](#)

Status for on-demand capture requests.

Enums

[CameraDevice.ErrorCode](#)

Error codes from the native plugin.

[CameraInfo.CameraEye](#)

The camera eye.

[CameraInfo.CameraSource](#)

The source of the camera feed.

[CaptureFailure.ErrorCode](#)

Error codes from Camera2.

[CaptureSessionBase<TProxy>.ErrorCode](#)

Error codes from the native plugin.

[CaptureTemplate](#)

Capture template to use when recording.

[PCASupport](#)

[ResourceState](#)

State of a Camera2 resource.

[StreamUseCase](#)

Stream Use Cases are a way to improve the performance of Camera2 capture sessions. They give the hardware device more information to tune parameters, which provides a better camera experience for your specific task.

Delegates

[CaptureSessionBase<TProxy>.ModifyRequestBuilderCallback](#)

A callback for modification of all [CaptureRequest builders](#) created for the session.

[CaptureSessionBase<TProxy>.ShouldRegisterCaptureEventsCallback](#)

A callback for configuring if capture-specific events should be registered for a capture request. Defaults to [false](#).

[ContinuousCaptureSession.OnFrameReadyCallback](#)

Processes a frame received from the native capture session.

Class CameraDevice

Namespace: [Uralstech.UXR.QuestCamera](#)

Manages a camera device.

```
public sealed class CameraDevice : StatefulResource
```

Inheritance

object ← [StatefulResource](#) ← CameraDevice

Inherited Members

[StatefulResource.State](#) , [StatefulResource.WaitForInitialization\(\)](#) ,
[StatefulResource.WaitForInitialization\(TimeSpan, Action, WaitTimeoutMode\)](#) ,
[StatefulResource.WaitForInitializationAsync\(CancellationToken\)](#).

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#) , [JNIExtensions.ToJava<T>\(T\)](#).

Constructors

CameraDevice(string)

```
public CameraDevice(string cameraId)
```

Parameters

cameraId string

Fields

CameraId

The ID of the camera.

```
public readonly string CameraId
```

Field Value

string

NativeProxy

Native callback handler.

```
public readonly CameraDevice.Proxy NativeProxy
```

Field Value

[CameraDevice.Proxy](#)

Methods

CreateContinuousPipeline(Resolution, CaptureTemplate, StreamUseCase, GraphicsFormat)

Creates a continuous capture pipeline (session + linked YUV to RGBA converter) for use.

```
public CapturePipeline<ContinuousCaptureSession>?  
CreateContinuousPipeline(Resolution resolution, CaptureTemplate template =  
CaptureTemplate.Preview, StreamUseCase streamUseCase = StreamUseCase.None,  
GraphicsFormat textureFormat = null)
```

Parameters

resolution Resolution

template [CaptureTemplate](#)

streamUseCase [StreamUseCase](#)

textureFormat GraphicsFormat

See [YUVConverter\(Resolution, ComputeShaderKernel, GraphicsFormat\)](#) for default texture format.

Returns

[CapturePipeline](#) <[ContinuousCaptureSession](#)>

The created pipeline, or [null](#) if creation failed.

CreateContinuousSession(Resolution, CaptureTemplate, StreamUseCase)

Creates a continuous capture session for use.

```
public ContinuousCaptureSession CreateContinuousSession(Resolution resolution,
CaptureTemplate template = CaptureTemplate.Preview, StreamUseCase streamUseCase
= StreamUseCase.None)
```

Parameters

resolution Resolution

The capture resolution. Must be from [SupportedResolutions](#).

template [CaptureTemplate](#)

The template to use for the captures.

streamUseCase [StreamUseCase](#)

The stream use case for this session. Must be from [SupportedStreamUseCases](#) or [None](#).

Returns

[ContinuousCaptureSession](#)

Returns the session. Check [State](#) (inherited by [ContinuousCaptureSession](#)) for the state of the session.

Remarks

To shut down and dispose the session, use [DisposeAsync\(\)](#) (inherited by [ContinuousCaptureSession](#)).

Exceptions

CreateGLESSessionAsync(Resolution, CaptureTemplate, StreamUseCase, GraphicsFormat)

Creates a generic OpenGL-ES based capture session.

```
public ValueTask<GLESCaptureSession> CreateGLESSessionAsync(Resolution resolution,
    CaptureTemplate template = CaptureTemplate.Preview, StreamUseCase streamUseCase =
    StreamUseCase.None, GraphicsFormat textureFormat = null)
```

Parameters

resolution Resolution

The capture resolution. Must be from [SupportedResolutions](#).

template [CaptureTemplate](#)

The template to use for the captures.

streamUseCase [StreamUseCase](#)

The stream use case for this session. Must be from [SupportedStreamUseCases](#) or [None](#).

textureFormat GraphicsFormat

The output texture format for the converted frames. See [GLESCaptureSession\(Resolution, GraphicsFormat\)](#) for default.

Returns

ValueTask<[GLESCaptureSession](#)>

Returns the session. Check [State](#) (inherited by [GLESCaptureSession](#)) for the state of the session.

Remarks

This initializes a native capture session backed by a SurfaceTexture and a GLES conversion job. The returned session must be started manually (e.g., via its run loop or single-run methods) and disposed using [DisposeAsync\(\)](#).

Exceptions

ObjectDisposedException

CreateOnDemandPipeline(Resolution, StreamUseCase, GraphicsFormat)

Creates an on-demand capture pipeline (session + linked YUV to RGBA converter) for use.

```
public CapturePipeline<OnDemandCaptureSession>? CreateOnDemandPipeline(Resolution resolution, StreamUseCase streamUseCase = StreamUseCase.None, GraphicsFormat textureFormat = null)
```

Parameters

resolution Resolution

streamUseCase [StreamUseCase](#)

textureFormat GraphicsFormat

See [YUVConverter\(Resolution, ComputeShaderKernel, GraphicsFormat\)](#) for default texture format.

Returns

[CapturePipeline](#)<[OnDemandCaptureSession](#)>

The created pipeline, or [null](#) if creation failed.

CreateOnDemandSession(Resolution, StreamUseCase)

Creates an on-demand capture session for use.

```
public OnDemandCaptureSession CreateOnDemandSession(Resolution resolution, StreamUseCase streamUseCase = StreamUseCase.None)
```

Parameters

resolution Resolution

The capture resolution. Must be from [SupportedResolutions](#).

streamUseCase [StreamUseCase](#)

The stream use case for this session. Must be from [SupportedStreamUseCases](#) or [None](#).

Returns

[OnDemandCaptureSession](#)

Returns the session. Check [State](#) (inherited by [OnDemandCaptureSession](#)) for the state of the session.

Remarks

To shut down and dispose the session, use [DisposeAsync\(\)](#) (inherited by [OnDemandCaptureSession](#)).

Exceptions

[ObjectDisposedException](#)

CreateVulkanContinuousSession(Resolution, CaptureTemplate, StreamUseCase, GraphicsFormat)

Creates a Vulkan-based continuous capture session for use.

```
public VulkanContinuousCaptureSession CreateVulkanContinuousSession(Resolution
resolution, CaptureTemplate template = CaptureTemplate.Preview, StreamUseCase
streamUseCase = StreamUseCase.None, GraphicsFormat textureFormat = null)
```

Parameters

resolution Resolution

The capture resolution. Must be from [SupportedResolutions](#).

template [CaptureTemplate](#)

The template to use for the captures.

streamUseCase [StreamUseCase](#)

The stream use case for this session. Must be from [SupportedStreamUseCases](#) or [None](#).

textureFormat [GraphicsFormat](#)

The output texture format for the converted frames. See [VulkanContinuousCaptureSession\(Resolution, GraphicsFormat\)](#) for default.

Returns

[VulkanContinuousCaptureSession](#)

Returns the session. Check [State](#) (inherited by [VulkanContinuousCaptureSession](#)) for the state of the session.

Remarks

To shut down and dispose the session, use [DisposeAsync\(\)](#).

Exceptions

[ObjectDisposedException](#)

CreateVulkanOnDemandSession(Resolution, CaptureTemplate, StreamUseCase, GraphicsFormat)

Creates a Vulkan-based on-demand capture session for use.

```
public VulkanOnDemandCaptureSession CreateVulkanOnDemandSession(Resolution
resolution, CaptureTemplate template = CaptureTemplate.Preview, StreamUseCase
streamUseCase = StreamUseCase.None, GraphicsFormat textureFormat = null)
```

Parameters

resolution [Resolution](#)

The capture resolution. Must be from [SupportedResolutions](#).

template [CaptureTemplate](#)

The template to use for the captures.

streamUseCase [StreamUseCase](#)

The stream use case for this session. Must be from [SupportedStreamUseCases](#) or [None](#).

textureFormat [GraphicsFormat](#)

The output texture format for the converted frames. See [VulkanOnDemandCaptureSession\(Resolution, GraphicsFormat\)](#) for default.

Returns

[VulkanOnDemandCaptureSession](#)

Returns the session. Check [State](#) (inherited by [VulkanOnDemandCaptureSession](#)) for the state of the session.

Remarks

To shut down and dispose the session, use [DisposeAsync\(\)](#) (inherited by [VulkanOnDemandCaptureSession](#)).

Exceptions

[ObjectDisposedException](#)

DisposeAsync()

Closes the camera (if not already closed) and releases native resources.

```
public ValueTask DisposeAsync()
```

Returns

[ValueTask](#)

~CameraDevice()

```
protected ~CameraDevice()
```

ThrowIfDisposed()

```
protected override void ThrowIfDisposed()
```

Events

OnDeviceClosed

Invoked when the camera is closed, along with the camera ID.

```
public event Action<string>? OnDeviceClosed
```

Event Type

Action<string>

OnDeviceDisconnected

Invoked when the camera is disconnected, along with the camera ID.

```
public event Action<string>? OnDeviceDisconnected
```

Event Type

Action<string>

OnDeviceErred

Invoked when the camera encounters an error, along with the camera ID.

```
public event Action<string, CameraDevice.ErrorCode>? OnDeviceErred
```

Event Type

Action<string, [CameraDevice.ErrorCode](#)>

OnDeviceOpened

Invoked when the camera is opened, along with the camera ID.

```
public event Action<string>? OnDeviceOpened
```

Event Type

Action<string>

Enum CameraDevice.ErrorCode

Namespace: [Uralstech.UXR.QuestCamera](#)

Error codes from the native plugin.

```
public enum CameraDevice.ErrorCode
```

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#), [JNIExtensions.ToJava<T>\(T\)](#)

Fields

CameraAccessError = 1001

The camera is disabled by device policy, has been disconnected, or is being used by a higher-priority camera API client.

CameraDeviceError = 4

The camera device has encountered a fatal error.

CameraDisabled = 3

The camera device could not be opened due to a device policy.

CameraInUse = 1

The camera device is in use already.

CameraServiceError = 5

The camera service has encountered a fatal error.

IllegalArgumentException = 1000

cameraId was null, or the cameraId does not match any currently or previously available camera device.

MaxCamerasInUse = 2

The camera device could not be opened because there are too many other open camera devices.

SecurityError = 1002

The application does not have permission to access the camera.

Unknown = 0

Class CameraDevice.Proxy

Namespace: [Uralstech.UXR.QuestCamera](#)

Java proxy to handle native callbacks.

```
public sealed class CameraDevice.Proxy : AndroidJavaProxy
```

Inheritance

object ← CameraDevice.Proxy

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#) , [JNIExtensions.ToJava<T>\(T\)](#)

Remarks

All event callbacks will be on a Java thread, and are performance sensitive.

Constructors

Proxy()

```
public Proxy()
```

Events

OnClosed

Invoked when the camera is closed.

```
public event Action? OnClosed
```

Event Type

Action

OnDisconnected

Invoked when the camera is disconnected.

```
public event Action? OnDisconnected
```

Event Type

Action

OnErred

Invoked when the camera encounters an error.

```
public event Action<CameraDevice.ErrorCode>? OnErred
```

Event Type

Action<[CameraDevice.ErrorCode](#)>

OnOpened

Invoked when the camera is opened.

```
public event Action? OnOpened
```

Event Type

Action

Class CameraInfo

Namespace: [Uralstech.UXR.QuestCamera](#)

Information describing a camera device.

```
public sealed class CameraInfo : CameraMetadata
```

Inheritance

object ← [CameraMetadata](#) ← CameraInfo

Inherited Members

[CameraMetadata.Native](#) , [CameraMetadata.KeyProviderClass](#) ,
[CameraMetadata.GetConstant\(string\)](#) , [CameraMetadata.GetKeys\(\)](#) ,
[CameraMetadata.TryGet<T>\(string, out T\)](#) ,
[CameraMetadata.TryGet<T>\(CameraMetadata.Key, out T\)](#) , [CameraMetadata.Dispose\(\)](#)

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#) , [JNIExtensions.ToJava<T>\(T\)](#)

Constructors

CameraInfo(string, AndroidJavaObject, Key?, Key?)

```
public CameraInfo(string cameraId, AndroidJavaObject native, CameraMetadata.Key?  
metaQuestSourceKey = null, CameraMetadata.Key? metaQuestPositionKey = null)
```

Parameters

cameraId string

native AndroidJavaObject

metaQuestSourceKey [CameraMetadata.Key](#)

metaQuestPositionKey [CameraMetadata.Key](#)

Fields

CameraId

The device ID of the camera.

```
public readonly string CameraId
```

Field Value

string

Eye

The eye the camera is closest to.

```
public readonly CameraInfo.CameraEye Eye
```

Field Value

[CameraInfo.CameraEye](#)

Intrinsics

The intrinsic data for this camera.

```
public readonly CameraInfo.CameraIntrinsics? Intrinsics
```

Field Value

[CameraInfo.CameraIntrinsics](#)

LensPoseRotation

The orientation of the camera relative to the sensor coordinate system, converted from Android sensor rotation space to Unity rotation space.

```
public readonly Quaternion? LensPoseRotation
```

Field Value

Quaternion?

LensPoseTranslation

The position of the camera's optical center, converted from Android sensor space to Unity space.

```
public readonly Vector3? LensPoseTranslation
```

Field Value

Vector3?

Source

The source of the camera feed.

```
public readonly CameraInfo.CameraSource Source
```

Field Value

[CameraInfo.CameraSource](#)

SupportedResolutions

The resolutions supported by this camera.

```
public readonly Resolution[] SupportedResolutions
```

Field Value

Resolution[]

SupportedStreamUseCases

The stream use cases supported by this camera, if any.

```
public readonly StreamUseCase[] SupportedStreamUseCases
```

Field Value

[StreamUseCase\[\]](#)

Remarks

Guaranteed to an empty array on devices < API Level 33.

Methods

Equals(object?)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object? obj)
```

Parameters

obj object

The object to compare with the current object.

Returns

bool

[true](#) if the specified object is equal to the current object; otherwise, [false](#).

Equals(CameraInfo?)

Indicates whether the current object is equal to another object of the same type.

```
public bool Equals(CameraInfo? other)
```

Parameters

other [CameraInfo](#)

An object to compare with this object.

Returns

bool

[true](#) if the current object is equal to the **other** parameter; otherwise, [false](#).

GetAvailableCaptureRequestKeys()

Returns the keys supported by this CameraDevice for querying with a CaptureRequest.

```
public CameraMetadata.Key[] GetAvailableCaptureRequestKeys()
```

Returns

[Key](#)[]

GetAvailableCaptureResultKeys()

Returns the keys supported by this CameraDevice for querying with a CaptureResult.

```
public CameraMetadata.Key[] GetAvailableCaptureResultKeys()
```

Returns

[Key](#)[]

GetAvailableSessionCharacteristicsKeys()

Returns the keys for this CameraDevice whose values are capture session specific.

```
public CameraMetadata.Key[] GetAvailableSessionCharacteristicsKeys()
```

Returns

[Key\[\]](#)

Remarks

Returns an empty array on devices < API Level 35.

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

GetKeysNeedingPermission()

Returns a subset of the keys returned by [GetKeys\(\)](#) with all keys that require camera clients to obtain the Manifest.permission.CAMERA permission.

```
public CameraMetadata.Key[] GetKeysNeedingPermission()
```

Returns

[Key\[\]](#)

ToString()

Returns a string that represents the current object.

```
public override string ToString()
```

Returns

string

A string that represents the current object.

Operators

operator ==(CameraInfo?, CameraInfo?)

```
public static bool operator ==(CameraInfo? a, CameraInfo? b)
```

Parameters

a [CameraInfo](#)

b [CameraInfo](#)

Returns

bool

operator !=(CameraInfo?, CameraInfo?)

```
public static bool operator !=(CameraInfo? a, CameraInfo? b)
```

Parameters

a [CameraInfo](#)

b [CameraInfo](#)

Returns

bool

Enum CameraInfo.CameraEye

Namespace: [Uralstech.UXR.QuestCamera](#)

The camera eye.

```
public enum CameraInfo.CameraEye
```

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#) , [JNIExtensions.ToJava<T>\(T\)](#).

Fields

Left = 0

Right = 1

Unknown = -1

Class CameraInfo.CameraIntrinsics

Namespace: [Uralstech.UXR.QuestCamera](#)

Defines the camera's intrinsic properties. All values are in pixels.

```
public sealed record CameraInfo.CameraIntrinsics
```

Inheritance

object $\leftarrow$ CameraInfo.CameraIntrinsics

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#) , [JNIExtensions.ToJava<T>\(T\)](#)

Constructors

CameraIntrinsics(Vector2, Vector2, Vector2, float)

Defines the camera's intrinsic properties. All values are in pixels.

```
public CameraIntrinsics(Vector2 Resolution, Vector2 FocalLength, Vector2  
PrincipalPoint, float Skew)
```

Parameters

Resolution Vector2

Resolution in pixels.

FocalLength Vector2

Focal length in pixels.

PrincipalPoint Vector2

Principal point in pixels from the image's top-left corner.

Skew float

Skew coefficient for axis misalignment.

Properties

FocalLength

Focal length in pixels.

```
public Vector2 FocalLength { get; init; }
```

Property Value

Vector2

PrincipalPoint

Principal point in pixels from the image's top-left corner.

```
public Vector2 PrincipalPoint { get; init; }
```

Property Value

Vector2

Resolution

Resolution in pixels.

```
public Vector2 Resolution { get; init; }
```

Property Value

Vector2

Skew

Skew coefficient for axis misalignment.

```
public float Skew { get; init; }
```

Property Value

float

Enum CameraInfo.CameraSource

Namespace: [Uralstech.UXR.QuestCamera](#)

The source of the camera feed.

```
public enum CameraInfo.CameraSource
```

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#) , [JNIExtensions.ToJava<T>\(T\)](#).

Fields

PassthroughRGB = 0

Unknown = -1

Class CameraMetadata

Namespace: [Uralstech.UXR.QuestCamera](#)

Represents a map of camera device, capture request or capture result metadata.

```
public abstract class CameraMetadata
```

Inheritance

object ← CameraMetadata

Derived

[CameraInfo](#), [CaptureRequest](#), [CaptureRequest.Builder](#), [CaptureResult](#)

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#), [JNIExtensions.ToJava<T>\(T\)](#)

Constructors

CameraMetadata(AndroidJavaObject, string)

```
public CameraMetadata(AndroidJavaObject native, string className)
```

Parameters

native AndroidJavaObject

className string

Fields

KeyProviderClass

The Java/Kotlin class which provides the keys and constants for this instance.

```
public readonly AndroidJavaClass KeyProviderClass
```

Field Value

AndroidJavaClass

Native

The native object.

```
public readonly AndroidJavaObject Native
```

Field Value

AndroidJavaObject

Methods

Dispose()

Performs application-defined tasks associated with freeing, releasing, or resetting unmanaged resources.

```
public void Dispose()
```

GetConstant(string)

Gets a constant value from [android.hardware.camera2.CameraMetadata](#).

```
public int GetConstant(string name)
```

Parameters

name string

The name of the constant field.

Returns

int

GetKeys()

Returns the keys contained in this map.

```
public CameraMetadata.Key[] GetKeys()
```

Returns

[Key\[\]](#)

Remarks

The caller is responsible for the disposal of all returned [CameraMetadata.Key](#) objects.

GetKeysFromListNative(string)

```
protected CameraMetadata.Key[] GetKeysFromListNative(string methodName)
```

Parameters

methodName string

Returns

[Key\[\]](#)

ThrowIfDisposed()

```
protected void ThrowIfDisposed()
```

TryGet<T>(string, out T)

Tries to get a value from this map.

```
public bool TryGet<T>(string keyName, out T value)
```

Parameters

keyName string

The name of the key.

value T

The result.

Returns

bool

[true](#) if the key exists and was converted successfully into a managed type, [false](#) otherwise.

Type Parameters

T

The type of the value.

Remarks

Supports converting to the types supported by [ToManaged<T>\(AndroidJavaObject\)](#) + `AndroidJavaObject`.

Exceptions

`KeyNotFoundException`

Thrown if the key is not defined in [KeyProviderClass](#).

TryGet<T>(Key, out T)

Tries to get a value from this map.

```
public bool TryGet<T>(CameraMetadata.Key key, out T value)
```

Parameters

key [CameraMetadata.Key](#)

The key.

value T

The result.

Returns

bool

[true](#) if the key exists and was converted successfully into a managed type, [false](#) otherwise.

Type Parameters

T

The type of the value.

Remarks

Supports converting to the types supported by [ToManaged<T>\(AndroidJavaObject\)](#) + [AndroidJavaObject](#).

Class CameraMetadata.FloatRange

Namespace: [Uralstech.UXR.QuestCamera](#)

A range of two float values.

```
public sealed record CameraMetadata.FloatRange
```

Inheritance

object ← CameraMetadata.FloatRange

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#) , [JNIExtensions.ToJava<T>\(T\)](#).

Constructors

FloatRange(float, float)

A range of two float values.

```
public FloatRange(float Lower, float Upper)
```

Parameters

Lower float

The lower endpoint.

Upper float

The upper endpoint.

Properties

Lower

```
public float Lower { get; init; }
```

Property Value

float

Upper

```
public float Upper { get; init; }
```

Property Value

float

Class CameraMetadata.IntRange

Namespace: [Uralstech.UXR.QuestCamera](#)

A range of two integer values.

```
public sealed record CameraMetadata.IntRange
```

Inheritance

object $\leftarrow$ CameraMetadata.IntRange

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#) , [JNIExtensions.ToJava<T>\(T\)](#).

Constructors

IntRange(int, int)

A range of two integer values.

```
public IntRange(int Lower, int Upper)
```

Parameters

Lower int

The lower endpoint.

Upper int

The upper endpoint.

Properties

Lower

The lower endpoint.


```
public int Lower { get; init; }
```

Property Value

int

Upper

The upper endpoint.

```
public int Upper { get; init; }
```

Property Value

int

Class CameraMetadata.Key

Namespace: [Uralstech.UXR.QuestCamera](#)

Light wrapper for [CaptureRequest.Key](#), [CameraCharacteristics.Key](#), etc.

```
public sealed class CameraMetadata.Key
```

Inheritance

object ← CameraMetadata.Key

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#), [JNIExtensions.ToJava<T>\(T\)](#)

Remarks

DO NOT mix or compare keys from different sources!

Constructors

Key(AndroidJavaObject)

```
public Key(AndroidJavaObject native)
```

Parameters

native AndroidJavaObject

Fields

Name

The name of the key.

```
public readonly string Name
```

Field Value

string

Native

The native object.

```
public readonly AndroidJavaObject Native
```

Field Value

AndroidJavaObject

Methods

Dispose()

Performs application-defined tasks associated with freeing, releasing, or resetting unmanaged resources.

```
public void Dispose()
```

Equals(object?)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object? obj)
```

Parameters

obj object

The object to compare with the current object.

Returns

bool

[true](#) if the specified object is equal to the current object; otherwise, [false](#).

Equals(Key?)

Indicates whether the current object is equal to another object of the same type.

```
public bool Equals(CameraMetadata.Key? other)
```

Parameters

other [CameraMetadata.Key](#)

An object to compare with this object.

Returns

bool

[true](#) if the current object is equal to the **other** parameter; otherwise, [false](#).

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

ToString()

Returns a string that represents the current object.

```
public override string ToString()
```

Returns

string

A string that represents the current object.

Operators

operator ==(Key?, Key?)

```
public static bool operator ==(CameraMetadata.Key? a, CameraMetadata.Key? b)
```

Parameters

a [CameraMetadata.Key](#)

b [CameraMetadata.Key](#)

Returns

bool

operator !=(Key?, Key?)

```
public static bool operator !=(CameraMetadata.Key? a, CameraMetadata.Key? b)
```

Parameters

a [CameraMetadata.Key](#)

b [CameraMetadata.Key](#)

Returns

bool

Class CameraMetadata.LongRange

Namespace: [Uralstech.UXR.QuestCamera](#)

A range of two long values.

```
public sealed record CameraMetadata.LongRange
```

Inheritance

object ← CameraMetadata.LongRange

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#) , [JNIExtensions.ToJava<T>\(T\)](#).

Constructors

LongRange(long, long)

A range of two long values.

```
public LongRange(long Lower, long Upper)
```

Parameters

Lower long

The lower endpoint.

Upper long

The upper endpoint.

Properties

Lower

```
public long Lower { get; init; }
```

Property Value

long

Upper

```
public long Upper { get; init; }
```

Property Value

long

Class CaptureFailure

Namespace: [Uralstech.UXR.QuestCamera](#)

A report of failed capture for a single image capture from the image sensor.

```
public sealed class CaptureFailure
```

Inheritance

object ← CaptureFailure

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#) , [JNIExtensions.ToJava<T>\(T\)](#).

Constructors

CaptureFailure(AndroidJavaObject)

```
public CaptureFailure(AndroidJavaObject native)
```

Parameters

native AndroidJavaObject

Fields

FrameNumber

The frame number associated with this failed capture.

```
public readonly long FrameNumber
```

Field Value

long

Native

The native object.

```
public readonly AndroidJavaObject Native
```

Field Value

AndroidJavaObject

Reason

The reason for the capture result being dropped.

```
public readonly CaptureFailure.ErrorCode Reason
```

Field Value

[CaptureFailure.ErrorCode](#)

SequenceId

The sequence ID of the capture request originally returned in [OnRequestSetWithId](#).

```
public readonly int SequenceId
```

Field Value

int

WasImageCaptured

If the image was captured from the camera.

```
public readonly bool WasImageCaptured
```

Field Value

bool

Methods

Dispose()

Performs application-defined tasks associated with freeing, releasing, or resetting unmanaged resources.

```
public void Dispose()
```

GetRequest()

Returns the capture request associated with this failure.

```
public CaptureRequest GetRequest()
```

Returns

[CaptureRequest](#)

Enum CaptureFailure.ErrorCode

Namespace: [Uralstech.UXR.QuestCamera](#)

Error codes from Camera2.

```
public enum CaptureFailure.ErrorCode
```

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#) , [JNIExtensions.ToJava<T>\(T\)](#).

Fields

FlushedError = 1

The capture has failed due to an abort call from the application.

FrameworkError = 0

The CaptureResult has been dropped this frame only due to an error in the framework.

Class CapturePipeline<T>

Namespace: [Uralstech.UXR.QuestCamera](#)

A wrapper for a complete capture-conversion pipeline.

```
public sealed class CapturePipeline<T> where T : ContinuousCaptureSession
```

Type Parameters

T

The capture session type.

Inheritance

object ← CapturePipeline<T>

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#) , [JNIExtensions.ToJava<T>\(T\)](#).

Constructors

CapturePipeline(T, YUVConverter)

```
public CapturePipeline(T session, YUVConverter converter)
```

Parameters

session T

converter [YUVConverter](#)

Fields

Converter

The YUV to RGBA texture converter.

```
public readonly YUVConverter Converter
```

Field Value

[YUVConverter](#)

Session

The capture session.

```
public readonly T Session
```

Field Value

T

Methods

DisposeAsync()

Performs application-defined tasks associated with freeing, releasing, or resetting unmanaged resources asynchronously.

```
public ValueTask DisposeAsync()
```

Returns

ValueTask

A task that represents the asynchronous dispose operation.

Class CaptureRequest

Namespace: [Uralstech.UXR.QuestCamera](#)

An immutable package of settings and outputs needed to capture a single image from the camera device.

```
public sealed class CaptureRequest : CameraMetadata
```

Inheritance

object ← [CameraMetadata](#) ← CaptureRequest

Inherited Members

[CameraMetadata.Native](#) , [CameraMetadata.KeyProviderClass](#) ,
[CameraMetadata.GetConstant\(string\)](#) , [CameraMetadata.GetKeys\(\)](#) ,
[CameraMetadata.TryGet<T>\(string, out T\)](#) ,
[CameraMetadata.TryGet<T>\(CameraMetadata.Key, out T\)](#) , [CameraMetadata.Dispose\(\)](#).

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#) , [JNIExtensions.ToJava<T>\(T\)](#).

Constructors

CaptureRequest(AndroidJavaObject)

```
public CaptureRequest(AndroidJavaObject native)
```

Parameters

native AndroidJavaObject

Methods

GetTag<T>()

Gets the tag of this request previously set using [SetTag<T>\(T\)](#).

```
public T? GetTag<T>()
```

Returns

T

Type Parameters

T

Remarks

Supports the types supported by [ToManaged<T>\(AndroidJavaObject\)](#) + AndroidJavaObject.

Class CaptureRequest.Builder

Namespace: [Uralstech.UXR.QuestCamera](#)

Builder for a capture request.

```
public sealed class CaptureRequest.Builder : CameraMetadata
```

Inheritance

object $\leftarrow$ [CameraMetadata](#) $\leftarrow$ CaptureRequest.Builder

Inherited Members

[CameraMetadata.Native](#) , [CameraMetadata.KeyProviderClass](#) ,
[CameraMetadata.GetConstant\(string\)](#) , [CameraMetadata.GetKeys\(\)](#) ,
[CameraMetadata.TryGet<T>\(string, out T\)](#) ,
[CameraMetadata.TryGet<T>\(CameraMetadata.Key, out T\)](#) , [CameraMetadata.Dispose\(\)](#)

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#) , [JNIExtensions.ToJava<T>\(T\)](#)

Constructors

Builder(AndroidJavaObject)

```
public Builder(AndroidJavaObject native)
```

Parameters

native AndroidJavaObject

Methods

SetTag<T>(T)

Set a tag for this request.

```
public void SetTag<T>(T tag) where T : notnull
```

Parameters

tag T

An arbitrary Object to store with this request.

Type Parameters

T

The type of the tag.

Remarks

Supports the types supported by [ToJava<T>\(T\)](#) + AndroidJavaObject.

Set<T>(string, T)

Set a capture request field to a value.

```
public void Set<T>(string keyName, T value) where T : notnull
```

Parameters

keyName string

The name of the key.

value T

The value to set.

Type Parameters

T

The type of the value.

Remarks

Supports the types supported by [ToJava<T>\(T\)](#) + AndroidJavaObject.

Exceptions

KeyNotFoundException

Thrown if the key is not defined in [KeyProviderClass](#).

Set<T>(Key, T)

Set a capture request field to a value.

```
public void Set<T>(CameraMetadata.Key key, T value) where T : notnull
```

Parameters

key [CameraMetadata.Key](#)

The key.

value T

The value to set.

Type Parameters

T

The type of the value.

Remarks

Supports the types supported by [ToJava<T>\(T\)](#) + AndroidJavaObject.

Class CaptureResult

Namespace: [Uralstech.UXR.QuestCamera](#)

The subset of the results of a single image capture from the image sensor.

```
public class CaptureResult : CameraMetadata
```

Inheritance

object ← [CameraMetadata](#) ← CaptureResult

Derived

[TotalCaptureResult](#)

Inherited Members

[CameraMetadata.Native](#) , [CameraMetadata.KeyProviderClass](#) ,
[CameraMetadata.GetConstant\(string\)](#) , [CameraMetadata.GetKeys\(\)](#) ,
[CameraMetadata.GetKeysFromListNative\(string\)](#) ,
[CameraMetadata.TryGet<T>\(string, out T\)](#) ,
[CameraMetadata.TryGet<T>\(CameraMetadata.Key, out T\)](#) , [CameraMetadata.Dispose\(\)](#) ,
[CameraMetadata.ThrowIfDisposed\(\)](#).

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#) , [JNIExtensions.ToJava<T>\(T\)](#).

Constructors

CaptureResult(AndroidJavaObject)

```
public CaptureResult(AndroidJavaObject native)
```

Parameters

native AndroidJavaObject

Methods

GetCameraId()

Returns the camera ID of the camera that produced this capture result.

```
public string GetCameraId()
```

Returns

string

GetFrameNumber()

Returns the frame number associated with this result.

```
public long GetFrameNumber()
```

Returns

long

GetRequest()

Returns the capture request associated with this result.

```
public CaptureRequest GetRequest()
```

Returns

[CaptureRequest](#)

GetSequenceId()

Returns the sequence ID of the capture request originally returned in [OnRequestSetWithId](#).

```
public int GetSequenceId()
```

Returns

int

Class CaptureSessionBase<TProxy>

Namespace: [Uralstech.UXR.QuestCamera](#)

Base class for all capture session types.

```
public abstract class CaptureSessionBase<TProxy> : StatefulResource where TProxy :  
    CaptureSessionBase<TProxy>.ProxyBase
```

Type Parameters

TProxy

The proxy type.

Inheritance

object ← [StatefulResource](#) ← CaptureSessionBase<TProxy>

Derived

[ContinuousCaptureSession](#), [GLESCaptureSession](#), [VulkanContinuousCaptureSession](#)

Inherited Members

[StatefulResource.State](#) , [StatefulResource.WaitForInitialization\(\)](#) ,
[StatefulResource.WaitForInitialization\(TimeSpan, Action, WaitTimeoutMode\)](#) ,
[StatefulResource.WaitForInitializationAsync\(CancellationToken\)](#)

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#) , [JNIExtensions.ToJava<T>\(T\)](#)

Constructors

CaptureSessionBase(TProxy, AndroidJavaObject)

```
protected CaptureSessionBase(TProxy proxy, AndroidJavaObject native)
```

Parameters

proxy TProxy

native AndroidJavaObject

Fields

NativeProxy

Native callback handler.

```
public readonly TProxy NativeProxy
```

Field Value

TProxy

_disposed

```
protected bool _disposed
```

Field Value

bool

_native

```
protected readonly AndroidJavaObject _native
```

Field Value

AndroidJavaObject

Methods

CloseWork()


```
protected ValueTask CloseWork()
```

Returns

ValueTask

DisposeAsync()

Closes the session (if not already closed) and releases native resources.

```
public virtual ValueTask DisposeAsync()
```

Returns

ValueTask

~CaptureSessionBase()

```
protected ~CaptureSessionBase()
```

MakeProxy<T>(out T)

```
protected static T MakeProxy<T>(out T proxy) where T :  
CaptureSessionBase<TProxy>.ProxyBase, new()
```

Parameters

proxy T

Returns

T

Type Parameters

ThrowIfDisposed()

```
protected override void ThrowIfDisposed()
```

TryAbortCaptures(out ErrorCode)

Discard all captures currently pending and in-progress as fast as possible.

```
public bool TryAbortCaptures(out CaptureSessionBase<TProxy>.ErrorCode errorCode)
```

Parameters

errorCode [CaptureSessionBase](#)<TProxy>.[ErrorCode](#)

Error code if the operation was unsuccessful.

Returns

bool

[true](#)  if successful; [false](#)  otherwise.

Remarks

This makes the session unusable for the future, so call [DisposeAsync\(\)](#) afterwards.

Exceptions

ObjectDisposedException

Events

OnSessionClosed

Called when the session is closed.

```
public event Action? OnSessionClosed
```

Event Type

Action

OnSessionConfigurationFailed

Called when the session could not be configured.

```
public event Action<CaptureSessionBase<TProxy>.ErrorCode>?  
OnSessionConfigurationFailed
```

Event Type

Action<[CaptureSessionBase](#)<TProxy>.[ErrorCode](#)>

OnSessionConfigured

Called when the session has been configured.

```
public event Action? OnSessionConfigured
```

Event Type

Action

OnSessionRequestFailed

Called when the session request could not be set.

```
public event Action<CaptureSessionBase<TProxy>.ErrorCode>? OnSessionRequestFailed
```

Event Type

Action<[CaptureSessionBase](#)<TProxy>.[ErrorCode](#)>

OnSessionRequestSet

Called when the session request has been set.

```
public event Action? OnSessionRequestSet
```

Event Type

Action

OnSessionRequestSetWithId

Same as [OnSessionRequestSet](#), but includes the sequence ID of the capture request.

```
public event Action<int>? OnSessionRequestSetWithId
```

Event Type

Action<int>

Enum

CaptureSessionBase<TProxy>.ErrorCode

Namespace: [Uralstech.UXR.QuestCamera](#)

Error codes from the native plugin.

```
public enum CaptureSessionBase<TProxy>.ErrorCode
```

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#) , [JNIExtensions.ToJava<T>\(T\)](#)

Fields

CameraAccessError = 1001

In case the camera device is no longer connected or has encountered a fatal error.

ConfigurationFailed = 2000

The session could not be configured.

IllegalArgumentError = 1000

In case the session configuration is invalid, if the captureTemplate is not supported by this device, or other illegal arguments were passed.

IllegalStateError = 1003

In case the camera device or session has been closed.

NativeJobBindingFailed = 3000

The Kotlin session could not bind to its C++ job.

OutOfResourcesError = 1004

If the surface or surfaceTexture for the session could not be created due to an out-of-resources error.

Delegate CaptureSessionBase<TProxy>.ModifyRequestBuilderCallback

Namespace: [Uralstech.UXR.QuestCamera](#)

A callback for modification of all [CaptureRequest builders](#)[↗] created for the session.

```
public delegate void  
CaptureSessionBase<TProxy>.ModifyRequestBuilderCallback(CaptureRequest.Builder  
builder, bool isRepeatingRequest)
```

Parameters

builder [CaptureRequest.Builder](#)

The builder object. DO NOT dispose this, or persist a reference to it beyond the callback.

isRepeatingRequest bool

If this builder is for a repeating or on-demand capture request.

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#), [JNIExtensions.ToJava<T>\(T\)](#)

Class

CaptureSessionBase<TProxy>.ProxyBase

Namespace: [Uralstech.UXR.QuestCamera](#)

Java proxy to handle native callbacks.

```
public abstract class CaptureSessionBase<TProxy>.ProxyBase : AndroidJavaProxy
```

Inheritance

object ← CaptureSessionBase<TProxy>.ProxyBase

Derived

[ContinuousCaptureSession.Proxy](#), [GLESCaptureSession.Proxy](#),
[VulkanContinuousCaptureSession.Proxy](#)

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#) , [JNIExtensions.ToJava<T>\(T\)](#)

Remarks

All event callbacks will be on a Java thread, and are performance sensitive.

Constructors

ProxyBase()

```
public ProxyBase()
```

ProxyBase(string)

```
protected ProxyBase(string className)
```

Parameters

className string

Events

ModifyRequestBuilder

A callback for modification of all [CaptureRequest builders](#) created for the session.

```
public event CaptureSessionBase<TProxy>.ModifyRequestBuilderCallback?  
ModifyRequestBuilder
```

Event Type

[CaptureSessionBase](#)<TProxy>.[ModifyRequestBuilderCallback](#)

Remarks

You should generally avoid configuring the request if `isRepeatingRequest` is [true](#) for on-demand sessions.

OnCaptureCompleted

Called when an image capture has fully completed and all the result metadata is available.

```
public event Action<CaptureRequest, TotalCaptureResult>? OnCaptureCompleted
```

Event Type

Action<[CaptureRequest](#), [TotalCaptureResult](#)>

Remarks

Override [ShouldRegisterCaptureEvents](#) for capture requests that should invoke this event. DO NOT persist references beyond the callback.

OnCaptureFailed

Called instead of [OnCaptureCompleted](#) when the camera device failed to produce a `CaptureResult` for the request.


```
public event Action<CaptureRequest, CaptureFailure>? OnCaptureFailed
```

Event Type

Action<[CaptureRequest](#), [CaptureFailure](#)>

Remarks

Override [ShouldRegisterCaptureEvents](#) for capture requests that should invoke this event. DO NOT persist references beyond the callback.

OnCaptureSequenceAborted

Called when a capture sequence aborts before any CaptureResult or CaptureFailure for it have been returned via this listener.

```
public event Action<int>? OnCaptureSequenceAborted
```

Event Type

Action<int>

Remarks

Override [ShouldRegisterCaptureEvents](#) for capture requests that should invoke this event. DO NOT persist references beyond the callback.

OnCaptureSequenceCompleted

Called when a capture sequence finishes and all CaptureResult or CaptureFailure for it have been returned via this listener.

```
public event Action<int, long>? OnCaptureSequenceCompleted
```

Event Type

Action<int, long>

Remarks

Override [ShouldRegisterCaptureEvents](#) for capture requests that should invoke this event. DO NOT persist references beyond the callback.

OnClosed

Called when the session is closed.

```
public event Action? OnClosed
```

Event Type

Action

OnConfigureFailed

Called when the session could not be configured.

```
public event Action<CaptureSessionBase<TProxy>.ErrorCode>? OnConfigureFailed
```

Event Type

Action<[CaptureSessionBase](#)<TProxy>.[ErrorCode](#)>

OnConfigured

Called when the session has been configured.

```
public event Action? OnConfigured
```

Event Type

Action

OnRequestFailed

Called when the session request could not be set.

```
public event Action<CaptureSessionBase<TProxy>.ErrorCode>? OnRequestFailed
```

Event Type

Action<[CaptureSessionBase](#)<TProxy>.[ErrorCode](#)>

OnRequestSet

Called when the session request has been set.

```
public event Action? OnRequestSet
```

Event Type

Action

OnRequestSetWithId

Same as [OnRequestSet](#), but includes the sequence ID of the capture request.

```
public event Action<int>? OnRequestSetWithId
```

Event Type

Action<int>

ShouldRegisterCaptureEvents

A callback for configuring if capture-specific events should be registered for a capture request. Defaults to [false](#) .

```
public event CaptureSessionBase<TProxy>.ShouldRegisterCaptureEventsCallback?  
ShouldRegisterCaptureEvents
```

Event Type

[CaptureSessionBase](#)<TProxy>.[ShouldRegisterCaptureEventsCallback](#)

Remarks

Please only register one listener to this event for each session.

Delegate CaptureSessionBase<TProxy>.ShouldRegisterCaptureEventsCallback

Namespace: [Uralstech.UXR.QuestCamera](#)

A callback for configuring if capture-specific events should be registered for a capture request. Defaults to [false](#) .

```
public delegate bool  
CaptureSessionBase<TProxy>.ShouldRegisterCaptureEventsCallback(CaptureRequest  
request, bool isRepeatingRequest)
```

Parameters

request [CaptureRequest](#)

The request for which events should be registered.

isRepeatingRequest bool

If this is for a repeating or on-demand capture request.

Returns

bool

A callback for configuring if capture-specific events should be registered for a capture request. Defaults to false.

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#) , [JNIExtensions.ToJava<T>\(T\)](#)

Remarks

Please only register one listener to this event for each session.

Enum CaptureTemplate

Namespace: [Uralstech.UXR.QuestCamera](#)

Capture template to use when recording.

```
public enum CaptureTemplate
```

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#) , [JNIExtensions.ToJava<T>\(T\)](#).

Fields

Preview = 1

Creates a request suitable for a camera preview window.

Record = 3

Creates a request suitable for video recording.

StillCapture = 2

Creates a request suitable for still image capture.

VideoSnapshot = 4

Creates a request suitable for still image capture while recording video.

Class ComputeShaderKernel

Namespace: [Uralstech.UXR.QuestCamera](#)

Utility class to contain ComputeShader kernel info.

```
[Serializable]  
public sealed class ComputeShaderKernel
```

Inheritance

object ← ComputeShaderKernel

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#), [JNIExtensions.ToJava<T>\(T\)](#)

Constructors

ComputeShaderKernel()

```
public ComputeShaderKernel()
```

ComputeShaderKernel(ComputeShader, string)

```
public ComputeShaderKernel(ComputeShader shader, string kernel = "CSMain")
```

Parameters

shader ComputeShader

kernel string

Properties

Index

```
public int Index { get; }
```

Property Value

int

Name

```
public string Name { get; }
```

Property Value

string

Shader

```
public ComputeShader Shader { get; }
```

Property Value

ComputeShader

ThreadGroupSizes

```
public Vector3Int ThreadGroupSizes { get; }
```

Property Value

Vector3Int

Methods

Validate()


```
public bool Validate()
```

Returns

bool

Class ContinuousCaptureSession

Namespace: [Uralstech.UXR.QuestCamera](#)

Manages a camera capture session with a continuous/repeating request.

```
public class ContinuousCaptureSession :  
    CaptureSessionBase<ContinuousCaptureSession.Proxy>
```

Inheritance

object ← [StatefulResource](#) ← [CaptureSessionBase](#)<[ContinuousCaptureSession.Proxy](#)> ←
ContinuousCaptureSession

Derived

[OnDemandCaptureSession](#)

Inherited Members

[CaptureSessionBase](#)<[ContinuousCaptureSession.Proxy](#)>.OnSessionConfigured ,
[CaptureSessionBase](#)<[ContinuousCaptureSession.Proxy](#)>.OnSessionConfigurationFailed ,
[CaptureSessionBase](#)<[ContinuousCaptureSession.Proxy](#)>.OnSessionRequestSet ,
[CaptureSessionBase](#)<[ContinuousCaptureSession.Proxy](#)>.OnSessionRequestSetWithId ,
[CaptureSessionBase](#)<[ContinuousCaptureSession.Proxy](#)>.OnSessionRequestFailed ,
[CaptureSessionBase](#)<[ContinuousCaptureSession.Proxy](#)>.OnSessionClosed ,
[CaptureSessionBase](#)<[ContinuousCaptureSession.Proxy](#)>.NativeProxy ,
[CaptureSessionBase](#)<[ContinuousCaptureSession.Proxy](#)>._native ,
[CaptureSessionBase](#)<[ContinuousCaptureSession.Proxy](#)>._disposed ,
[CaptureSessionBase](#)<[ContinuousCaptureSession.Proxy](#)>.MakeProxy<T>(out T) ,
[CaptureSessionBase](#)<[ContinuousCaptureSession.Proxy](#)>.TryAbortCaptures(out
[CaptureSessionBase](#)<[ContinuousCaptureSession.Proxy](#)>.ErrorCode) ,
[CaptureSessionBase](#)<[ContinuousCaptureSession.Proxy](#)>.DisposeAsync() ,
[CaptureSessionBase](#)<[ContinuousCaptureSession.Proxy](#)>.CloseWork() ,
[CaptureSessionBase](#)<[ContinuousCaptureSession.Proxy](#)>.ThrowIfDisposed() ,
[StatefulResource.State](#) , [StatefulResource.WaitForInitialization\(\)](#) ,
[StatefulResource.WaitForInitialization\(TimeSpan, Action, WaitTimeoutMode\)](#) ,
[StatefulResource.WaitForInitializationAsync\(CancellationToken\)](#)

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#) , [JNIExtensions.ToJava<T>\(T\)](#)

Remarks

This does not do any image processing/conversion on its own, but classes like [YUVConverter](#) can register callbacks to [OnFrameReady](#) to do their own processing of the raw image YUV 4:2:0, BT.601 Full Range data.

Constructors

ContinuousCaptureSession(Resolution)

```
public ContinuousCaptureSession(Resolution resolution)
```

Parameters

resolution Resolution

ContinuousCaptureSession(Resolution, string)

```
protected ContinuousCaptureSession(Resolution resolution, string className)
```

Parameters

resolution Resolution

className string

Delegate ContinuousCaptureSession.OnFrameReady Callback

Namespace: [Uralstech.UXR.QuestCamera](#)

Processes a frame received from the native capture session.

```
public delegate void ContinuousCaptureSession.OnFrameReadyCallback(nint yBuffer,  
long yBufferSize, nint uBuffer, nint vBuffer, long uvBufferSize, int yRowStride, int  
uvRowStride, int uvPixelStride, long timestamp)
```

Parameters

yBuffer nint

The pointer to this frame's Y (luminance) data.

yBufferSize long

The size of the Y buffer in bytes.

uBuffer nint

The pointer to this frame's U (color) data.

vBuffer nint

The pointer to this frame's V (color) data.

uvBufferSize long

The size of the U and V buffers in bytes.

yRowStride int

The size of each row of the image in **yBuffer** in bytes.

uvRowStride int

The size of each row of the image in **uBuffer** and **vBuffer** in bytes.

`uvPixelStride` int

The size of a pixel in a row of the image in `uBuffer` and `vBuffer` in bytes.

`timestamp` long

The timestamp the frame was captured at in nanoseconds.

Extension Methods

[`JNIExtensions.ToJava\(object, Type\)`](#) , [`JNIExtensions.ToJava<T>\(T\)`](#).

Class ContinuousCaptureSession.Proxy

Namespace: [Uralstech.UXR.QuestCamera](#)

Java proxy to handle native callbacks.

```
public sealed class ContinuousCaptureSession.Proxy :  
    CaptureSessionBase<ContinuousCaptureSession.Proxy>.ProxyBase
```

Inheritance

object ← [CaptureSessionBase<ContinuousCaptureSession.Proxy>.ProxyBase](#) ←
ContinuousCaptureSession.Proxy

Inherited Members

[CaptureSessionBase<ContinuousCaptureSession.Proxy>.ProxyBase.ModifyRequestBuilder](#) ,
[CaptureSessionBase<ContinuousCaptureSession.Proxy>.ProxyBase.ShouldRegisterCaptureEvents](#) ,
[CaptureSessionBase<ContinuousCaptureSession.Proxy>.ProxyBase.OnConfigured](#) ,
[CaptureSessionBase<ContinuousCaptureSession.Proxy>.ProxyBase.OnConfigureFailed](#) ,
[CaptureSessionBase<ContinuousCaptureSession.Proxy>.ProxyBase.OnRequestSet](#) ,
[CaptureSessionBase<ContinuousCaptureSession.Proxy>.ProxyBase.OnRequestSetWithId](#) ,
[CaptureSessionBase<ContinuousCaptureSession.Proxy>.ProxyBase.OnRequestFailed](#) ,
[CaptureSessionBase<ContinuousCaptureSession.Proxy>.ProxyBase.OnCaptureCompleted](#) ,
[CaptureSessionBase<ContinuousCaptureSession.Proxy>.ProxyBase.OnCaptureFailed](#) ,
[CaptureSessionBase<ContinuousCaptureSession.Proxy>.ProxyBase.OnCaptureSequenceCompleted](#)
,
[CaptureSessionBase<ContinuousCaptureSession.Proxy>.ProxyBase.OnCaptureSequenceAborted](#) ,
[CaptureSessionBase<ContinuousCaptureSession.Proxy>.ProxyBase.OnClosed](#)

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#) , [JNIExtensions.ToJava<T>\(T\)](#).

Remarks

All event callbacks will be on a Java thread, and are performance sensitive.

Constructors

Proxy()

```
public Proxy()
```

Methods

Invoke(string, nint)

```
public override nint Invoke(string methodName, nint javaArgs)
```

Parameters

methodName string

javaArgs nint

Returns

nint

Events

OnFrameReady

Processes a frame received from the native capture session.

```
public event ContinuousCaptureSession.OnFrameReadyCallback? OnFrameReady
```

Event Type

[ContinuousCaptureSession.OnFrameReadyCallback](#)

Remarks

See [ContinuousCaptureSession.OnFrameReadyCallback](#) for parameters.

Class JNIExtensions

Namespace: [Uralstech.UXR.QuestCamera](#)

QOL extensions for the JNI.

```
public static class JNIExtensions
```

Inheritance

object ← JNIExtensions

Remarks

This class is [public](#) to allow package users to reuse these extensions if useful. However, it should not be considered stable and is not part of the supported public API. It exists solely as an internal utility and may change or be removed at any time.

Methods

JavaListAsManagedArray<T>(AndroidJavaObject)

Converts a java.util.List to a managed array.

```
public static T[] JavaListAsManagedArray<T>(this AndroidJavaObject current)
```

Parameters

current AndroidJavaObject

Returns

T[]

The converted array.

Type Parameters

T

The type of the elements.

Remarks

WARNING: This method **DOES NOT** verify if the native Java/Kotlin elements can actually be converted to the target type.

Supports converting elements to the types supported by [ToManaged<T>](#) ([AndroidJavaObject](#)) + `AndroidJavaObject`.

ToJava(object, Type)

Converts managed .NET/Unity object to a Java/Kotlin object represented by an `AndroidJavaObject`.

```
[SuppressMessage("Style", "IDE0046:Convert to conditional expression", Justification = "Code is neater without conditionals.")]  
public static AndroidJavaObject ToJava(this object current, Type target)
```

Parameters

current object

target Type

The type of the managed object.

Returns

`AndroidJavaObject`

The converted Java object.

Remarks

Basic supported types:

- `bool`
- `char`
- `sbyte`
- `short`
- `int`

- long
- float
- double
- string

Supported Android types:

Unity Type	Android Type
Resolution	android.util.Size 
Rect	android.graphics.RectF 
RectInt	android.graphics.Rect 
CameraMetadata.IntRange	android.util.Range<Integer> 
CameraMetadata.LongRange	android.util.Range<Long>
CameraMetadata.FloatRange	android.util.Range<Float>

Also supports converting arrays of all above types + generic `AndroidJavaObject[]` and `AndroidJavaProxy[]`.

Exceptions

NotSupportedException

Thrown when the type is unsupported.

ToJava<T>(T)

Converts managed .NET/Unity object to a Java/Kotlin object represented by an `AndroidJavaObject`.

```
public static AndroidJavaObject ToJava<T>(this T current) where T : notnull
```

Parameters

current T

Returns

AndroidJavaObject

The converted Java object.

Type Parameters

T

The type of the managed object.

Remarks

Basic supported types:

- bool
- char
- sbyte
- short
- int
- long
- float
- double
- string

Supported Android types:

Unity Type	Android Type
Resolution	android.util.Size 
Rect	android.graphics.RectF 
RectInt	android.graphics.Rect 
CameraMetadata.IntRange	android.util.Range<Integer> 
CameraMetadata.LongRange	android.util.Range<Long>
CameraMetadata.FloatRange	android.util.Range<Float>

Also supports converting arrays of all above types + generic AndroidJavaObject[] and AndroidJavaProxy[].

Exceptions

NotSupportedException

Thrown when the type is unsupported.

ToManaged(AndroidJavaObject, Type)

Converts a Java/Kotlin object represented by an AndroidJavaObject into a managed .NET/Unity type.

```
[SuppressMessage("Style", "IDE0046:Convert to conditional expression", Justification = "Code is neater without conditionals.")]  
public static object ToManaged(this AndroidJavaObject current, Type target)
```

Parameters

current AndroidJavaObject

target Type

The managed target type.

Returns

object

The converted managed object.

Remarks

WARNING: This method **DOES NOT** verify if the native Java/Kotlin object can actually be converted to the target type.

Basic supported types:

- bool
- char
- sbyte
- short
- int
- long

- float
- double
- string

Supported Android types:

Unity Type	Android Type
Resolution	android.util.Size 
Rect	android.graphics.RectF 
RectInt	android.graphics.Rect 
CameraMetadata.IntRange	android.util.Range<Integer> 
CameraMetadata.LongRange	android.util.Range<Long>
CameraMetadata.FloatRange	android.util.Range<Float>

Also supports converting arrays of all above types + generic `AndroidJavaObject[]`.

Exceptions

NotSupportedException

Thrown when the requested target type is unsupported.

ToManaged<T>(AndroidJavaObject)

```
public static T ToManaged<T>(this AndroidJavaObject current)
```

Parameters

current AndroidJavaObject

Returns

T

Type Parameters

T

The managed target type.

TryConvertToManaged<T>(AndroidJavaObject, out T?)

Safe version of [ToManaged<T>\(AndroidJavaObject\)](#).

```
public static bool TryConvertToManaged<T>(this AndroidJavaObject current, out
T? value)
```

Parameters

current AndroidJavaObject

value T

Returns

bool

Type Parameters

T

UnboxBoolElement(nint, int)

Unboxes a boolean from a native Object array.

```
public static bool UnboxBoolElement(nint args, int index)
```

Parameters

args nint

The native array to take the boolean from.

index int

The index of the boolean object in the native array.

Returns

bool

The unboxed boolean.

UnboxByteBufferElement(nint, int)

Retrieves a ByteBuffer object from a native object array and obtains its direct memory address.

```
public static (nint obj, nint ptr) UnboxByteBufferElement(nint args, int index)
```

Parameters

args nint

JNI object array containing the ByteBuffer.

index int

Index of the ByteBuffer within the array.

Returns

(nint obj, nint ptr)

A tuple containing the local JNI reference to the ByteBuffer, and the pointer to its direct buffer memory.

UnboxIntElement(nint, int)

Unboxes an integer from a native Object array.

```
public static int UnboxIntElement(nint args, int index)
```

Parameters

args nint

The native array to take the integer from.

index int

The index of the integer object in the native array.

Returns

int

The unboxed integer.

UnboxLongElement(nint, int)

Unboxes a long from a native Object array.

```
public static long UnboxLongElement(nint args, int index)
```

Parameters

args nint

The native array to take the long from.

index int

The index of the long object in the native array.

Returns

long

The unboxed long.

UnboxObjectElement(nint, int)

Unboxes an AndroidJavaObject from a native Object array.


```
public static AndroidJavaObject UnboxObjectElement(nint args, int index)
```

Parameters

args nint

The native array to take the object from.

index int

The index of the object in the native array.

Returns

AndroidJavaObject

The unboxed object.

Class OnDemandCaptureSession

Namespace: [Uralstech.UXR.QuestCamera](#)

Manages an on-demand, explicit capture session.

```
public sealed class OnDemandCaptureSession : ContinuousCaptureSession
```

Inheritance

object ← [StatefulResource](#) ← [CaptureSessionBase](#)<[ContinuousCaptureSession.Proxy](#)> ← [ContinuousCaptureSession](#) ← OnDemandCaptureSession

Inherited Members

[CaptureSessionBase](#)<[ContinuousCaptureSession.Proxy](#)>.OnSessionConfigured ,
[CaptureSessionBase](#)<[ContinuousCaptureSession.Proxy](#)>.OnSessionConfigurationFailed ,
[CaptureSessionBase](#)<[ContinuousCaptureSession.Proxy](#)>.OnSessionRequestSet ,
[CaptureSessionBase](#)<[ContinuousCaptureSession.Proxy](#)>.OnSessionRequestSetWithId ,
[CaptureSessionBase](#)<[ContinuousCaptureSession.Proxy](#)>.OnSessionRequestFailed ,
[CaptureSessionBase](#)<[ContinuousCaptureSession.Proxy](#)>.OnSessionClosed ,
[CaptureSessionBase](#)<[ContinuousCaptureSession.Proxy](#)>.NativeProxy ,
[CaptureSessionBase](#)<[ContinuousCaptureSession.Proxy](#)>.TryAbortCaptures(out [CaptureSessionBase](#)<[ContinuousCaptureSession.Proxy](#)>.ErrorCode) ,
[CaptureSessionBase](#)<[ContinuousCaptureSession.Proxy](#)>.DisposeAsync() ,
[StatefulResource.State](#) , [StatefulResource.WaitForInitialization\(\)](#) ,
[StatefulResource.WaitForInitialization\(TimeSpan, Action, WaitTimeoutMode\)](#) ,
[StatefulResource.WaitForInitializationAsync\(CancellationToken\)](#).

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#) , [JNIExtensions.ToJava<T>\(T\)](#).

Remarks

This does not do any image processing/conversion on its own, but classes like [YUVConverter](#) can register callbacks to [OnFrameReady](#) to do their own processing of the raw image YUV 4:2:0, BT.601 Full Range data.

Constructors

OnDemandCaptureSession(Resolution)

```
public OnDemandCaptureSession(Resolution resolution)
```

Parameters

resolution Resolution

Methods

RequestCapture(CaptureTemplate)

Requests a new capture from the session.

```
public OnDemandCaptureSession.RequestStatus RequestCapture(CaptureTemplate template  
= CaptureTemplate.StillCapture)
```

Parameters

template [CaptureTemplate](#)

Returns

[OnDemandCaptureSession.RequestStatus](#)

Exceptions

ObjectDisposedException

TryRequestCapture(out ErrorCode, CaptureTemplate)

Requests a new capture from the session.

```
public bool TryRequestCapture(out  
CaptureSessionBase<ContinuousCaptureSession.Proxy>.ErrorCode errorCode,  
CaptureTemplate template = CaptureTemplate.StillCapture)
```

Parameters

errorCode [CaptureSessionBase](#)<[ContinuousCaptureSession.Proxy](#)>.[ErrorCode](#)

Error code if the operation was unsuccessful.

template [CaptureTemplate](#)

The capture template to use for the capture.

Returns

bool

[true](#)  if successful; [false](#)  otherwise.

Exceptions

ObjectDisposedException

Struct

OnDemandCaptureSession.RequestStatus

Namespace: [Uralstech.UXR.QuestCamera](#)

Status for on-demand capture requests.

```
public readonly struct OnDemandCaptureSession.RequestStatus
```

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#) , [JNIExtensions.ToJava<T>\(T\)](#)

Constructors

RequestStatus(ErrorCode, int, bool)

```
public RequestStatus(CaptureSessionBase<ContinuousCaptureSession.Proxy>.ErrorCode  
errorCode, int sequenceId, bool success)
```

Parameters

errorCode [CaptureSessionBase<ContinuousCaptureSession.Proxy>.ErrorCode](#)

sequenceId int

success bool

Fields

ErrorCode

Error code for failures, only valid if [Success](#) is [false](#).

```
public readonly CaptureSessionBase<ContinuousCaptureSession.Proxy>.ErrorCode  
ErrorCode
```

Field Value

[CaptureSessionBase](#)<[ContinuousCaptureSession.Proxy](#)>.[ErrorCode](#)

SequenceId

Sequence ID of capture request, only valid if [Success](#) is [true](#)✓.

```
public readonly int SequenceId
```

Field Value

int

Success

If the request was successful.

```
public readonly bool Success
```

Field Value

bool

Enum PCASupport

Namespace: [Uralstech.UXR.QuestCamera](#)

```
public enum PCASupport
```

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#) , [JNIExtensions.ToJava<T>\(T\)](#)

Fields

Supported = 1

PCA API is supported.

Meta XR Core SDK on Quest 3/3S with Horizon OS v74+.

Unknown = 0

Support status cannot be determined.

Returned when the Meta XR Core SDK is not available.

Unsupported = 2

PCA API is not supported.

Non-Android device, or Meta XR Core SDK on unsupported devices or OS versions.

Class QuestCameraManager

Namespace: [Uralstech.UXR.QuestCamera](#)

Entry point for the native Camera2 plugin.

```
public sealed class QuestCameraManager : DontCreateNewSingleton<QuestCameraManager>
```

Inheritance

object ← QuestCameraManager

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#) , [JNIExtensions.ToJava<T>\(T\)](#).

Fields

AvatarCameraPermission

Meta Quest Avatar Camera API permission string.

```
public const string AvatarCameraPermission = "android.permission.CAMERA"
```

Field Value

string

ConversionKernel

The shader and kernel to use for YUV 4:2:0 to RGBA conversion.

```
public ComputeShaderKernel ConversionKernel
```

Field Value

[ComputeShaderKernel](#)

HeadsetCameraPermission

Meta Quest Passthrough Camera API permission string.

```
public const string HeadsetCameraPermission = "horizonos.permission.HEADSET_CAMERA"
```

Field Value

string

MetaQuestCameraPositionKeyName

Name of the metadata key for the Quest's camera eye position.

```
public const string MetaQuestCameraPositionKeyName =  
"com.meta.extra_metadata.position"
```

Field Value

string

MetaQuestCameraSourceKeyName

Name of the metadata key for the Quest's camera source.

```
public const string MetaQuestCameraSourceKeyName =  
"com.meta.extra_metadata.camera_source"
```

Field Value

string

Properties

Cameras

A managed, cached array of available cameras and their characteristics.

```
public IReadOnlyList<CameraInfo> Cameras { get; }
```

Property Value

`IReadOnlyList<CameraInfo>`

MetaQuestCameraPositionKey

Metadata key for the Quest's camera eye position.

```
public CameraMetadata.Key MetaQuestCameraPositionKey { get; }
```

Property Value

[CameraMetadata.Key](#)

MetaQuestCameraSourceKey

Metadata key for the Quest's camera source.

```
public CameraMetadata.Key MetaQuestCameraSourceKey { get; }
```

Property Value

[CameraMetadata.Key](#)

Support

Tries to get the runtime's support for the Passthrough Camera Access and Camera2 APIs.

```
public static PCASupport Support { get; }
```

Property Value

[PCASupport](#)

Methods

Awake()

```
protected override void Awake()
```

GetDevices()

Gets the IDs and intrinsics of all connected camera devices.

```
public CameraInfo[] GetDevices()
```

Returns

[CameraInfo](#)[]

OpenCamera(string)

Opens a camera device for use.

```
public CameraDevice OpenCamera(string cameraId)
```

Parameters

cameraId string

The ID of the camera to open.

Returns

[CameraDevice](#)

The camera device. Check [State](#) (inherited by [CameraDevice](#)) for the state of the device.

Remarks

Once you have finished using the camera, close and dispose of it using [DisposeAsync\(\)](#).

OpenCamera(CameraInfo)

Opens a camera device for use.

```
public CameraDevice OpenCamera(CameraInfo cameraInfo)
```

Parameters

cameraInfo [CameraInfo](#)

Returns

[CameraDevice](#)

The camera device. Check [State](#) (inherited by [CameraDevice](#)) for the state of the device.

Remarks

Once you have finished using the camera, close and dispose of it using [DisposeAsync\(\)](#).

RefreshDevices()

Refreshes cached camera device information.

```
public void RefreshDevices()
```

TryGetDevice(CameraEye, out CameraInfo?)

Finds a camera device by its corresponding eye.

```
public bool TryGetDevice(CameraInfo.CameraEye eye, out CameraInfo? cameraInfo)
```

Parameters

eye [CameraInfo.CameraEye](#)

cameraInfo [CameraInfo](#)

Returns

bool

Enum ResourceState

Namespace: [Uralstech.UXR.QuestCamera](#)

State of a Camera2 resource.

```
public enum ResourceState
```

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#) , [JNIExtensions.ToJava<T>\(T\)](#).

Fields

Initializing = 0

Resource is initializing.

Invalid = 2

Resource failed with an error, was disconnected or is being/was closed normally.

Valid = 1

Resource is open and ready.

Class StatefulResource

Namespace: [Uralstech.UXR.QuestCamera](#)

```
public abstract class StatefulResource
```

Inheritance

object ← StatefulResource

Derived

[CameraDevice](#), [CaptureSessionBase<TProxy>](#)

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#), [JNIExtensions.ToJava<T>\(T\)](#)

Properties

State

The state of this resource.

```
public ResourceState State { get; protected set; }
```

Property Value

[ResourceState](#)

Remarks

This does NOT indicate disposal status, and is just an indicator for if you can use this resource.

Methods

ThrowIfDisposed()

```
protected abstract void ThrowIfDisposed()
```

WaitForInitialization()

Waits until the resource opens or errs out.

```
public WaitUntil WaitForInitialization()
```

Returns

WaitUntil

Exceptions

ObjectDisposedException

WaitForInitialization(TimeSpan, Action, WaitTimeoutMode)

Waits until the resource opens or errs out.

```
public WaitUntil WaitForInitialization(TimeSpan timeout, Action onTimeout,
WaitTimeoutMode timeoutMode = null)
```

Parameters

timeout TimeSpan

Maximum time to wait.

onTimeout Action

The action to perform when the **timeout** is reached.

timeoutMode WaitTimeoutMode

Mode in which to measure time to determine **timeout**.

Returns

WaitUntil

Exceptions

ObjectDisposedException

WaitForInitializationAsync(CancellationTokens)

Waits until the resource opens or errs out.

```
public Task<bool> WaitForInitializationAsync(CancellationTokens token = default)
```

Parameters

token CancellationTokens

Returns

Task<bool>

[true](#) if opened successfully, [false](#) otherwise.

Exceptions

ObjectDisposedException

Enum StreamUseCase

Namespace: [Uralstech.UXR.QuestCamera](#)

Stream Use Cases are a way to improve the performance of Camera2 capture sessions. They give the hardware device more information to tune parameters, which provides a better camera experience for your specific task.

```
public enum StreamUseCase : long
```

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#) , [JNIExtensions.ToJava<T>\(T\)](#).

Fields

Default = 0

Default stream use case.

None = -1

Does not set any stream use case. Use this if the device does not support stream use cases or you're unsure of support.

Preview = 1

Live stream shown to the user.

PreviewVideoStill = 4

One single stream used for combined purposes of preview, video, and still capture.

StillCapture = 2

Still photo capture.

VideoCall = 5

Long-running video call optimized for both power efficiency and video quality.

VideoRecord = 3

Recording video clips.

Class TotalCaptureResult

Namespace: [Uralstech.UXR.QuestCamera](#)

The total assembled results of a single image capture from the image sensor.

```
public sealed class TotalCaptureResult : CaptureResult
```

Inheritance

object ← [CameraMetadata](#) ← [CaptureResult](#) ← TotalCaptureResult

Inherited Members

[CaptureResult.GetCameraId\(\)](#) , [CaptureResult.GetFrameNumber\(\)](#) ,
[CaptureResult.GetRequest\(\)](#) , [CaptureResult.GetSequenceId\(\)](#) , [CameraMetadata.Native](#) ,
[CameraMetadata.KeyProviderClass](#) , [CameraMetadata.GetConstant\(string\)](#) ,
[CameraMetadata.GetKeys\(\)](#) , [CameraMetadata.TryGet<T>\(string, out T\)](#) ,
[CameraMetadata.TryGet<T>\(CameraMetadata.Key, out T\)](#) , [CameraMetadata.Dispose\(\)](#)

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#) , [JNIExtensions.ToJava<T>\(T\)](#)

Constructors

TotalCaptureResult(AndroidJavaObject)

```
public TotalCaptureResult(AndroidJavaObject native)
```

Parameters

native AndroidJavaObject

Methods

GetPartialResults()

Returns the list of partial results that compose this total result.

```
public CaptureResult[] GetPartialResults()
```

Returns

[CaptureResult\[\]](#)

Class YUVConverter

Namespace: [Uralstech.UXR.QuestCamera](#)

Converts raw camera capture session frames to Unity-supported RGBA.

```
public sealed class YUVConverter
```

Inheritance

object ← YUVConverter

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#) , [JNIExtensions.ToJava<T>\(T\)](#)

Constructors

YUVConverter(Resolution, GraphicsFormat)

Creates a new converter with the shader and kernel described in the scene instance of [QuestCameraManager](#).

```
public YUVConverter(Resolution resolution, GraphicsFormat textureFormat = null)
```

Parameters

resolution Resolution

textureFormat GraphicsFormat

If not specified, uses equivalent of RenderTextureFormat.ARGB32.

YUVConverter(Resolution, ComputeShaderKernel, GraphicsFormat)

```
public YUVConverter(Resolution resolution, ComputeShaderKernel kernel,  
GraphicsFormat textureFormat = null)
```

Parameters

resolution Resolution

kernel [ComputeShaderKernel](#)

textureFormat GraphicsFormat

If not specified, uses equivalent of RenderTextureFormat.ARGB32.

Fields

Texture

The output texture with converted frames.

```
public readonly RenderTexture Texture
```

Field Value

RenderTexture

Properties

CaptureTimestamp

The capture timestamp of the last processed frame.

```
public long CaptureTimestamp { get; }
```

Property Value

long

HasNewFrame

[true](#) if a capture was processed this frame; [false](#) otherwise.

```
public bool HasNewFrame { get; }
```

Property Value

bool

ShaderKernel

The shader used for conversion.

```
public ComputeShaderKernel ShaderKernel { get; set; }
```

Property Value

[ComputeShaderKernel](#)

Methods

Dispose()

Performs application-defined tasks associated with freeing, releasing, or resetting unmanaged resources.

```
public void Dispose()
```

OnFrameReady(nint, long, nint, nint, long, int, int, int, long)

Processes a frame received from the native capture session.

```
public void OnFrameReady(nint yBuffer, long yBufferSize, nint uBuffer, nint vBuffer, long uvBufferSize, int yRowStride, int uvRowStride, int uvPixelStride, long timestamp)
```

Parameters

yBuffer nint

The pointer to this frame's Y (luminance) data.

yBufferSize long

The size of the Y buffer in bytes.

uBuffer nint

The pointer to this frame's U (color) data.

vBuffer nint

The pointer to this frame's V (color) data.

uvBufferSize long

The size of the U and V buffers in bytes.

yRowStride int

The size of each row of the image in **yBuffer** in bytes.

uvRowStride int

The size of each row of the image in **uBuffer** and **vBuffer** in bytes.

uvPixelStride int

The size of a pixel in a row of the image in **uBuffer** and **vBuffer** in bytes.

timestamp long

The timestamp the frame was captured at in nanoseconds.

Events

OnFrameProcessed

Callback for when a frame has been dispatched for conversion, with the frame texture and capture timestamp.

```
public event Action<RenderTexture, long>? OnFrameProcessed
```


Event Type

Action<RenderTexture, long>

Namespace Uralstech.UXR.QuestCamera. GLES

Classes

[GLESAPI](#)

Exposes the native GLES Texture Conversion API.

[GLESCaptureSession](#)

Manages a camera capture session with a repeating request, but supports both repeating and on-demand conversion.

[GLESCaptureSession.Proxy](#)

Java proxy to handle native callbacks.

Structs

[RenderJobDisposeData](#)

Data for [Dispose](#).

[RenderJobRunData](#)

Data for [Run](#).

[RenderJobSetupData](#)

Data for [Setup](#).

Enums

[RenderJobEvent](#)

IDs of all Render Job events.

Delegates

[RenderJobDisposeData.Callback](#)

Callback for when the job is disposed or the process fails.

[RenderJobRunData.Callback](#)

Callback for when the job finishes rendering or the process fails.

[RenderJobSetupData.Callback](#)

Callback for when the job is setup or the process fails.

Class GLESAPI

Namespace: [Uralstech.UXR.QuestCamera.GLES](#)

Exposes the native GLES Texture Conversion API.

```
public static class GLESAPI
```

Inheritance

object ← GLESAPI

Fields

DisposeCallbacksRegistry

Registry of job disposal callbacks. This is a single-call registry, i.e. the entry is removed after the callback occurs.

```
public static readonly ConcurrentDictionary<uint, RenderJobDisposeData.Callback>  
DisposeCallbacksRegistry
```

Field Value

ConcurrentDictionary<uint, [RenderJobDisposeData.Callback](#)>

RenderJobDisposeCallbackPtr

Static marshalled pointer to [OnRenderJobDispose\(bool, uint\)](#).

```
public static readonly nint RenderJobDisposeCallbackPtr
```

Field Value

nint

RenderJobRunCallbackPtr

Static marshalled pointer to [OnRenderJobRun\(long, uint\)](#).

```
public static readonly nint RenderJobRunCallbackPtr
```

Field Value

nint

RenderJobSetupCallbackPtr

Static marshalled pointer to [OnRenderJobSetup\(uint, uint\)](#).

```
public static readonly nint RenderJobSetupCallbackPtr
```

Field Value

nint

RunCallbacksRegistry

Registry of job run callbacks.

```
public static readonly ConcurrentDictionary<uint, RenderJobRunData.Callback>  
RunCallbacksRegistry
```

Field Value

ConcurrentDictionary<uint, [RenderJobRunData.Callback](#)>

SetupCallbacksRegistry

Registry of job setup callbacks. This is a single-call registry, i.e. the entry is removed after the callback occurs.

```
public static readonly ConcurrentDictionary<uint, RenderJobSetupData.Callback>  
SetupCallbacksRegistry
```

Field Value

ConcurrentDictionary<uint, [RenderJobSetupData.Callback](#)>

Methods

OnRenderJobDispose(bool, uint)

Callback for when the job is disposed or the process fails.

```
public static void OnRenderJobDispose(bool result, uint renderTextureId)
```

Parameters

result bool

Whether the operation was a success or not.

renderTextureId uint

[RenderTextureId](#), for lookup.

OnRenderJobRun(long, uint)

Callback for when the job finishes rendering or the process fails.

```
public static void OnRenderJobRun(long timestamp, uint renderTextureId)
```

Parameters

timestamp long

The timestamp returned by the SurfaceTexture, or -1 if the operation failed.

renderTextureId uint

[RenderTextureId](#), for lookup.

OnRenderJobSetup(uint, uint)

Callback for when the job is setup or the process fails.

```
public static void OnRenderJobSetup(uint nativeTextureId, uint renderTextureId)
```

Parameters

nativeTextureId uint

renderTextureId uint

[RenderTextureId](#), for lookup.

getGLESTextureManagerJobEvent()

Returns a pointer to the native render job management function.

```
public static extern nint getGLESTextureManagerJobEvent()
```

Returns

nint

Class GLESCaptureSession

Namespace: [Uralstech.UXR.QuestCamera.GLES](#)

Manages a camera capture session with a repeating request, but supports both repeating and on-demand conversion.

```
public sealed class GLESCaptureSession :  
    CaptureSessionBase<GLESCaptureSession.Proxy>
```

Inheritance

object ← [StatefulResource](#) ← [CaptureSessionBase](#)<[GLESCaptureSession.Proxy](#)> ←
GLESCaptureSession

Inherited Members

[CaptureSessionBase](#)<[GLESCaptureSession.Proxy](#)>.OnSessionConfigured ,
[CaptureSessionBase](#)<[GLESCaptureSession.Proxy](#)>.OnSessionConfigurationFailed ,
[CaptureSessionBase](#)<[GLESCaptureSession.Proxy](#)>.OnSessionRequestSet ,
[CaptureSessionBase](#)<[GLESCaptureSession.Proxy](#)>.OnSessionRequestSetWithId ,
[CaptureSessionBase](#)<[GLESCaptureSession.Proxy](#)>.OnSessionRequestFailed ,
[CaptureSessionBase](#)<[GLESCaptureSession.Proxy](#)>.OnSessionClosed ,
[CaptureSessionBase](#)<[GLESCaptureSession.Proxy](#)>.NativeProxy ,
[CaptureSessionBase](#)<[GLESCaptureSession.Proxy](#)>.TryAbortCaptures(out
[CaptureSessionBase](#)<[GLESCaptureSession.Proxy](#)>.ErrorCode) ,
[StatefulResource.State](#) , [StatefulResource.WaitForInitialization\(\)](#) ,
[StatefulResource.WaitForInitialization\(TimeSpan, Action, WaitTimeoutMode\)](#) ,
[StatefulResource.WaitForInitializationAsync\(CancellationToken\)](#).

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#) , [JNIExtensions.ToJava<T>\(T\)](#).

Remarks

Texture conversion is done through a native OpenGL-ES plugin.

Constructors

GLESCaptureSession(Resolution, GraphicsFormat)

```
public GLESCaptureSession(Resolution resolution, GraphicsFormat textureFormat  
= null)
```

Parameters

resolution Resolution

textureFormat GraphicsFormat

If not specified, uses equivalent of RenderTextureFormat.ARGB32.

Exceptions

NotSupportedException

Thrown if this constructor is invoked in an environment where OpenGL ES 3 is unavailable.

Fields

Texture

The output texture with converted frames.

```
public readonly Texture2D Texture
```

Field Value

Texture2D

Properties

CaptureTimestamp

The capture timestamp of the last processed frame.

```
public long CaptureTimestamp { get; }
```


Property Value

long

HasNewFrame

[true](#) if a capture was processed this frame; [false](#) otherwise.

```
public bool HasNewFrame { get; }
```

Property Value

bool

Methods

DisposeAsync()

Closes the session (if not already closed) and releases native resources.

```
public override ValueTask DisposeAsync()
```

Returns

ValueTask

ProcessSingleFrameAsync(CancellationToken)

Processes a single frame and returns the result.

```
public ValueTask<(long, Texture2D)> ProcessSingleFrameAsync(CancellationToken token = default)
```

Parameters

token CancellationToken

Returns

ValueTask<(long, Texture2D)>

Capture timestamp and updated texture. Timestamp will be -1 if the capture could not be processed.

Exceptions

InvalidOperationException

Thrown if continuous processing is active.

ObjectDisposedException

TimeoutException

SetupJobAsync()

Registers the texture and creates a job in the native C++ manager.

```
public ValueTask<uint> SetupJobAsync()
```

Returns

ValueTask<uint>

The ID of the source texture created for the job, which the capture session will render to, or 0 if the operation failed.

StartContinuousProcessing(int)

Starts continuous frame processing.

```
public void StartContinuousProcessing(int maxFramerate = 60)
```

Parameters

maxFramerate int

The maximum rate at which frames will be processed by the GLES pipeline.

Exceptions

InvalidOperationException

Thrown if continuous processing is already active.

ObjectDisposedException

Events

OnFrameProcessed

Callback for when a frame has been processed, with the frame texture and capture timestamp.

```
public event Action<Texture2D, long>? OnFrameProcessed
```

Event Type

Action<Texture2D, long>

Remarks

The image at this point has not *actually* finished processing, but all GPU commands to do so have been executed.

Class GLESCaptureSession.Proxy

Namespace: [Uralstech.UXR.QuestCamera.GLES](#)

Java proxy to handle native callbacks.

```
public sealed class GLESCaptureSession.Proxy :  
    CaptureSessionBase<GLESCaptureSession.Proxy>.ProxyBase
```

Inheritance

object ← [CaptureSessionBase<GLESCaptureSession.Proxy>.ProxyBase](#) ←
GLESCaptureSession.Proxy

Inherited Members

[CaptureSessionBase<GLESCaptureSession.Proxy>.ProxyBase.ModifyRequestBuilder](#) ,
[CaptureSessionBase<GLESCaptureSession.Proxy>.ProxyBase.ShouldRegisterCaptureEvents](#) ,
[CaptureSessionBase<GLESCaptureSession.Proxy>.ProxyBase.OnConfigured](#) ,
[CaptureSessionBase<GLESCaptureSession.Proxy>.ProxyBase.OnConfigureFailed](#) ,
[CaptureSessionBase<GLESCaptureSession.Proxy>.ProxyBase.OnRequestSet](#) ,
[CaptureSessionBase<GLESCaptureSession.Proxy>.ProxyBase.OnRequestSetWithId](#) ,
[CaptureSessionBase<GLESCaptureSession.Proxy>.ProxyBase.OnRequestFailed](#) ,
[CaptureSessionBase<GLESCaptureSession.Proxy>.ProxyBase.OnCaptureCompleted](#) ,
[CaptureSessionBase<GLESCaptureSession.Proxy>.ProxyBase.OnCaptureFailed](#) ,
[CaptureSessionBase<GLESCaptureSession.Proxy>.ProxyBase.OnCaptureSequenceCompleted](#)
,
[CaptureSessionBase<GLESCaptureSession.Proxy>.ProxyBase.OnCaptureSequenceAborted](#) ,
[CaptureSessionBase<GLESCaptureSession.Proxy>.ProxyBase.OnClosed](#)

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#) , [JNIExtensions.ToJava<T>\(T\)](#)

Remarks

All event callbacks will be on a Java thread, and are performance sensitive.

Struct RenderJobDisposeData

Namespace: [Uralstech.UXR.QuestCamera.GLES](#)

Data for [Dispose](#).

```
public readonly struct RenderJobDisposeData
```

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#) , [JNIExtensions.ToJava<T>\(T\)](#).

Constructors

RenderJobDisposeData(uint, nint)

```
public RenderJobDisposeData(uint renderTextureId, nint onDone)
```

Parameters

renderTextureId uint

onDone nint

Fields

OnDone

Method with signature of [RenderJobDisposeData.Callback](#).

```
public readonly nint OnDone
```

Field Value

nint

RenderTextureId

The Job's ID and render target.

```
public readonly uint RenderTextureId
```

Field Value

uint

Delegate RenderJobDisposeData.Callback

Namespace: [Uralstech.UXR.QuestCamera.GLES](#)

Callback for when the job is disposed or the process fails.

```
public delegate void RenderJobDisposeData.Callback(bool result,  
uint renderTextureId)
```

Parameters

result bool

Whether the operation was a success or not.

renderTextureId uint

[RenderTextureId](#), for lookup.

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#) , [JNIExtensions.ToJava<T>\(T\)](#).

Enum RenderJobEvent

Namespace: [Uralstech.UXR.QuestCamera.GLES](#)

IDs of all Render Job events.

```
public enum RenderJobEvent
```

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#) , [JNIExtensions.ToJava<T>\(T\)](#).

Fields

Dispose = 2

Disposes the native texture and cleans up the registered job.

Run = 3

Runs a job.

Setup = 1

Sets up a native texture to convert from, and creates a registerable job for the capture session.

Struct RenderJobRunData

Namespace: [Uralstech.UXR.QuestCamera.GLES](#)

Data for [Run](#).

```
public readonly struct RenderJobRunData
```

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#) , [JNIExtensions.ToJava<T>\(T\)](#).

Constructors

RenderJobRunData(uint, nint)

```
public RenderJobRunData(uint renderTextureId, nint onDone)
```

Parameters

renderTextureId uint

onDone nint

Fields

OnDone

Method with signature of [RenderJobRunData.Callback](#).

```
public readonly nint OnDone
```

Field Value

nint

RenderTextureId

The Job's ID and render target.

```
public readonly uint RenderTextureId
```

Field Value

uint

Delegate RenderJobRunData.Callback

Namespace: [Uralstech.UXR.QuestCamera.GLES](#)

Callback for when the job finishes rendering or the process fails.

```
public delegate void RenderJobRunData.Callback(long timestamp, uint renderTextureId)
```

Parameters

timestamp long

The timestamp returned by the SurfaceTexture, or -1 if the operation failed.

renderTextureId uint

[RenderTextureId](#), for lookup.

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#) , [JNIExtensions.ToJava<T>\(T\)](#)

Struct RenderJobSetupData

Namespace: [Uralstech.UXR.QuestCamera.GLES](#)

Data for [Setup](#).

```
public readonly struct RenderJobSetupData
```

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#) , [JNIExtensions.ToJava<T>\(T\)](#).

Constructors

RenderJobSetupData(uint, int, int, nint)

```
public RenderJobSetupData(uint renderTextureId, int width, int height, nint onDone)
```

Parameters

renderTextureId uint

width int

height int

onDone nint

Fields

Height

The height of [RenderTextureId](#).

```
public readonly int Height
```

Field Value

int

OnDone

Method with signature of [RenderJobSetupData.Callback](#).

```
public readonly nint OnDone
```

Field Value

nint

RenderTextureId

The GLES ID of the texture which will be the Job's ID and render target.

```
public readonly uint RenderTextureId
```

Field Value

uint

Width

The width of [RenderTextureId](#).

```
public readonly int Width
```

Field Value

int

Delegate RenderJobSetupData.Callback

Namespace: [Uralstech.UXR.QuestCamera.GLES](#)

Callback for when the job is setup or the process fails.

```
public delegate void RenderJobSetupData.Callback(uint nativeTexture,
uint renderTextureId)
```

Parameters

nativeTexture uint

The created texture, or 0 if the operation failed.

renderTextureId uint

[RenderTextureId](#), for lookup.

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#) , [JNIExtensions.ToJava<T>\(T\)](#).

Namespace Uralstech.UXR.QuestCamera.Vulkan

Classes

[VulkanAPI](#)

Exposes the native Vulkan Texture Conversion API.

[VulkanContinuousCaptureSession](#)

Manages a camera capture session with a continuous/repeating request.

[VulkanContinuousCaptureSession.Proxy](#)

Java proxy to handle native callbacks.

[VulkanOnDemandCaptureSession](#)

Manages a continuous capture session with user-invoked texture conversion.

Structs

[ManagedRenderCallback](#)

A managed callback for a [VulkanAPI](#) render event.

[RenderData](#)

Data for a [VulkanAPI](#) render event.

Delegates

[RenderData.Callback](#)

Callback for when the event completes.

[VulkanContinuousCaptureSession.OnFrameReadyCallback](#)

Signals the Vulkan plugin that a frame is ready for conversion.

Struct ManagedRenderCallback

Namespace: [Uralstech.UXR.QuestCamera.Vulkan](#)

A managed callback for a [VulkanAPI](#) render event.

```
public readonly struct ManagedRenderCallback
```

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#) , [JNIExtensions.ToJava<T>\(T\)](#).

Constructors

ManagedRenderCallback(nint, long, Action<bool, long>)

```
public ManagedRenderCallback(nint renderDataMemory, long timestampNs, Action<bool, long> onDone)
```

Parameters

renderDataMemory nint

timestampNs long

onDone Action<bool, long>

Fields

OnDone

The managed callback (bool Success, long [TimestampNs](#)).

```
public readonly Action<bool, long> OnDone
```

Field Value

Action<bool, long>

Remarks

This will be invoked from the render thread, do not call Unity APIs from it.

RenderDataMemory

The native memory allocated for event data.

```
public readonly nint RenderDataMemory
```

Field Value

nint

TimestampNs

The timestamp of the frame being rendered, in nanoseconds.

```
public readonly long TimestampNs
```

Field Value

long

Struct RenderData

Namespace: [Uralstech.UXR.QuestCamera.Vulkan](#)

Data for a [VulkanAPI](#) render event.

```
public readonly struct RenderData
```

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#) , [JNIExtensions.ToJava<T>\(T\)](#).

Constructors

RenderData(nint, int, ulong, nint, nint)

```
public RenderData(nint sourceHardwareBuffer, int sourceHardwareBufferDataSpace,
ulong sourceHardwareBufferId, nint destinationImage, nint onDone)
```

Parameters

sourceHardwareBuffer nint

sourceHardwareBufferDataSpace int

sourceHardwareBufferId ulong

destinationImage nint

onDone nint

Fields

Align

Alignment of this struct ([UnsafeUtility.AlignOf<T>\(\)](#)).

```
public static readonly int Align
```

Field Value

int

DestinationImage

The target Unity-created VkImage to render to.

```
public readonly nint DestinationImage
```

Field Value

nint

OnDone

Method with signature of [RenderData.Callback](#).

```
public readonly nint OnDone
```

Field Value

nint

Size

Size of this struct (UnsafeUtility.SizeOf<T>()).

```
public static readonly int Size
```

Field Value

int

SourceHardwareBuffer

The source AHardwareBuffer to render from.

```
public readonly nint SourceHardwareBuffer
```

Field Value

nint

SourceHardwareBufferDataSpace

The data space of [SourceHardwareBuffer](#).

```
public readonly int SourceHardwareBufferDataSpace
```

Field Value

int

SourceHardwareBufferId

The global unique ID of [SourceHardwareBuffer](#).

```
public readonly ulong SourceHardwareBufferId
```

Field Value

ulong

Delegate RenderData.Callback

Namespace: [Uralstech.UXR.QuestCamera.Vulkan](#)

Callback for when the event completes.

```
public delegate void RenderData.Callback(byte success, ulong hardwareBufferId)
```

Parameters

success byte

Was the event executed successfully?

hardwareBufferId ulong

[SourceHardwareBufferId](#), for lookup.

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#) , [JNIExtensions.ToJava<T>\(T\)](#)

Class VulkanAPI

Namespace: [Uralstech.UXR.QuestCamera.Vulkan](#)

Exposes the native Vulkan Texture Conversion API.

```
public static class VulkanAPI
```

Inheritance

object ← VulkanAPI

Fields

RenderCallbackPtr

Static marshalled pointer to [OnRenderDone\(byte, ulong\)](#).

```
public static readonly nint RenderCallbackPtr
```

Field Value

nint

RenderCallbacksRegistry

Registry of render callbacks.

```
public static readonly ConcurrentDictionary<long, ManagedRenderCallback>  
RenderCallbacksRegistry
```

Field Value

ConcurrentDictionary<long, [ManagedRenderCallback](#)>

Methods

AllocateRenderData(ref RenderData)

Allocates native memory of a [RenderData](#) struct to use with [OnRenderDone\(byte, ulong\)/FreeRenderData\(in nint\)](#).

```
public static nint AllocateRenderData(ref RenderData renderData)
```

Parameters

renderData [RenderData](#)

The data to allocate.

Returns

nint

A pointer to the allocated memory.

Remarks

Allocations are made with UnsafeUtility and Allocator.TempJob, matching de-allocation in [OnRenderDone\(byte, ulong\)](#) and [FreeRenderData\(in nint\)](#).

FreeRenderData(in nint)

Frees the memory of a [RenderData](#) struct allocated using [AllocateRenderData\(ref RenderData\)](#).

```
public static void FreeRenderData(in nint renderDataPtr)
```

Parameters

renderDataPtr **nint**

The memory to free.

OnRenderDone(byte, ulong)

Callback for when the event completes.

```
public static void OnRenderDone(byte success, ulong hardwareBufferId)
```

Parameters

success byte

Was the event executed successfully?

hardwareBufferId ulong

[SourceHardwareBufferId](#), for lookup.

getVulkanRenderEvent()

Returns a pointer to the native rendering function.

```
public static extern nint getVulkanRenderEvent()
```

Returns

nint

releaseHardwareBuffer(nint)

Releases an acquired AHardwareBuffer object.

```
public static extern void releaseHardwareBuffer(nint acquiredBufferPtr)
```

Parameters

acquiredBufferPtr nint

The buffer to release.

Class VulkanContinuousCaptureSession

Namespace: [Uralstech.UXR.QuestCamera.Vulkan](#)

Manages a camera capture session with a continuous/repeating request.

```
public class VulkanContinuousCaptureSession :  
    CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>
```

Inheritance

object ← [StatefulResource](#) ←
[CaptureSessionBase](#)<[VulkanContinuousCaptureSession.Proxy](#)> ←
VulkanContinuousCaptureSession

Derived

[VulkanOnDemandCaptureSession](#)

Inherited Members

[CaptureSessionBase](#)<[VulkanContinuousCaptureSession.Proxy](#)>.OnSessionConfigured ,
[CaptureSessionBase](#)<[VulkanContinuousCaptureSession.Proxy](#)>.OnSessionConfigurationFailed
,
[CaptureSessionBase](#)<[VulkanContinuousCaptureSession.Proxy](#)>.OnSessionRequestSet ,
[CaptureSessionBase](#)<[VulkanContinuousCaptureSession.Proxy](#)>.OnSessionRequestSetWithId ,
[CaptureSessionBase](#)<[VulkanContinuousCaptureSession.Proxy](#)>.OnSessionRequestFailed ,
[CaptureSessionBase](#)<[VulkanContinuousCaptureSession.Proxy](#)>.OnSessionClosed ,
[CaptureSessionBase](#)<[VulkanContinuousCaptureSession.Proxy](#)>.NativeProxy ,
[CaptureSessionBase](#)<[VulkanContinuousCaptureSession.Proxy](#)>._native ,
[CaptureSessionBase](#)<[VulkanContinuousCaptureSession.Proxy](#)>._disposed ,
[CaptureSessionBase](#)<[VulkanContinuousCaptureSession.Proxy](#)>.MakeProxy<T>(out T) ,
[CaptureSessionBase](#)<[VulkanContinuousCaptureSession.Proxy](#)>.TryAbortCaptures(out
[CaptureSessionBase](#)<[VulkanContinuousCaptureSession.Proxy](#)>.ErrorCode) ,
[CaptureSessionBase](#)<[VulkanContinuousCaptureSession.Proxy](#)>.CloseWork() ,
[CaptureSessionBase](#)<[VulkanContinuousCaptureSession.Proxy](#)>.ThrowIfDisposed() ,
[StatefulResource.State](#) , [StatefulResource.WaitForInitialization\(\)](#) ,
[StatefulResource.WaitForInitialization\(TimeSpan, Action, WaitTimeoutMode\)](#) ,
[StatefulResource.WaitForInitializationAsync\(CancellationToken\)](#)

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#) , [JNIExtensions.ToJava<T>\(T\)](#)

Remarks

Texture conversion is done through a native Vulkan plugin.

Constructors

VulkanContinuousCaptureSession(Resolution, GraphicsFormat)

```
public VulkanContinuousCaptureSession(Resolution resolution, GraphicsFormat textureFormat = null)
```

Parameters

resolution Resolution

textureFormat GraphicsFormat

If not specified, uses equivalent of RenderTextureFormat.ARGB32.

Exceptions

NotSupportedException

Thrown if this constructor is invoked in an environment where Vulkan is unavailable or if the current runtime's Android API Level is lower than 33 (Android 13).

Fields

Texture

The output texture with converted frames.

```
public readonly RenderTexture Texture
```

Field Value

RenderTexture

Properties

CaptureTimestamp

The capture timestamp of the last processed frame.

```
public long CaptureTimestamp { get; }
```

Property Value

long

HasNewFrame

[true](#) if a capture was processed this frame; [false](#) otherwise.

```
public bool HasNewFrame { get; }
```

Property Value

bool

Methods

DispatchFrameConversionAsync(nint, int, long, long)

```
protected Task DispatchFrameConversionAsync(nint acquiredBufferPtr, int  
bufferDataSpace, long bufferId, long timestampNs)
```

Parameters

acquiredBufferPtr nint

bufferDataSpace int

bufferId long

timestampNs long

Returns

Task

DisposeAsync()

Closes the session (if not already closed) and releases native resources.

```
public override ValueTask DisposeAsync()
```

Returns

ValueTask

OnFrameReadyNative(nint, int, long, long)

```
protected virtual void OnFrameReadyNative(nint acquiredBufferPtr, int bufferDataSpace,  
long bufferId, long timestampNs)
```

Parameters

acquiredBufferPtr nint

bufferDataSpace int

bufferId long

timestampNs long

Events

OnFrameProcessed

Callback for when a frame has been processed, with the frame texture and capture timestamp.

```
public event Action<RenderTexture, long>? OnFrameProcessed
```

Event Type

Action<RenderTexture, long>

Remarks

The image at this point has not *actually* finished processing, but all GPU commands to do so have been enqueued.

Delegate VulkanContinuousCaptureSession.OnFrameReadyCallback

Namespace: [Uralstech.UXR.QuestCamera.Vulkan](#)

Signals the Vulkan plugin that a frame is ready for conversion.

```
public delegate void VulkanContinuousCaptureSession.OnFrameReadyCallback(nint acquiredBufferPtr, int bufferDataSpace, long bufferId, long timestampNs)
```

Parameters

acquiredBufferPtr nint

The HardwareBuffer associated with the frame.

bufferDataSpace int

The [DataSpace](#) of the buffer.

bufferId long

The global unique ID ([AHardwareBuffer_getId](#)) of the buffer.

timestampNs long

The timestamp the frame was captured at in nanoseconds.

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#) , [JNIExtensions.ToJava<T>\(T\)](#).

Remarks

[AHardwareBuffer_acquire](#) has already been called on **acquiredBufferPtr**, and this method **must** release it either through native plugin render events or [releaseHardwareBuffer\(nint\)](#).

Class VulkanContinuousCaptureSession.Proxy

Namespace: [Uralstech.UXR.QuestCamera.Vulkan](#)

Java proxy to handle native callbacks.

```
public sealed class VulkanContinuousCaptureSession.Proxy :  
    CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>.ProxyBase
```

Inheritance

object ← [CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>.ProxyBase](#) ←
VulkanContinuousCaptureSession.Proxy

Inherited Members

[CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>.ProxyBase.ModifyRequestBuilder](#) ,
[CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>.ProxyBase.ShouldRegisterCaptureEvents](#) ,
[CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>.ProxyBase.OnConfigured](#) ,
[CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>.ProxyBase.OnConfigureFailed](#) ,
[CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>.ProxyBase.OnRequestSet](#) ,
[CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>.ProxyBase.OnRequestSetWithId](#) ,
[CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>.ProxyBase.OnRequestFailed](#) ,
[CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>.ProxyBase.OnCaptureCompleted](#) ,
[CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>.ProxyBase.OnCaptureFailed](#) ,
[CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>.ProxyBase.OnCaptureSequenceCompleted](#)
,
[CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>.ProxyBase.OnCaptureSequenceAborted](#) ,
[CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>.ProxyBase.OnClosed](#)

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#) , [JNIExtensions.ToJava<T>\(T\)](#)

Remarks

All event callbacks will be on a Java thread, and are performance sensitive.

Constructors

Proxy()

```
public Proxy()
```

Methods

Invoke(string, nint)

```
public override nint Invoke(string methodName, nint javaArgs)
```

Parameters

methodName string

javaArgs nint

Returns

nint

Events

OnFrameReady

Signals the Vulkan plugin that a frame is ready for conversion.

```
public event VulkanContinuousCaptureSession.OnFrameReadyCallback? OnFrameReady
```

Event Type

[VulkanContinuousCaptureSession.OnFrameReadyCallback](#)

Remarks

[AHardwareBuffer_acquire](#) has already been called on **acquiredBufferPtr**, and this method **must** release it either through native plugin render events or [releaseHardwareBuffer\(nint\)](#).

Class VulkanOnDemandCaptureSession

Namespace: [Uralstech.UXR.QuestCamera.Vulkan](#)

Manages a continuous capture session with user-invoked texture conversion.

```
public sealed class VulkanOnDemandCaptureSession : VulkanContinuousCaptureSession
```

Inheritance

object ← [StatefulResource](#) ←

[CaptureSessionBase](#)<[VulkanContinuousCaptureSession.Proxy](#)> ←

[VulkanContinuousCaptureSession](#) ← VulkanOnDemandCaptureSession

Inherited Members

[VulkanContinuousCaptureSession.OnFrameProcessed](#) ,

[VulkanContinuousCaptureSession.HasNewFrame](#) , [VulkanContinuousCaptureSession.Texture](#) ,

[VulkanContinuousCaptureSession.CaptureTimestamp](#) ,

[VulkanContinuousCaptureSession.DisposeAsync\(\)](#) ,

[CaptureSessionBase](#)<[VulkanContinuousCaptureSession.Proxy](#)>.OnSessionConfigured ,

[CaptureSessionBase](#)<[VulkanContinuousCaptureSession.Proxy](#)>.OnSessionConfigurationFailed

,

[CaptureSessionBase](#)<[VulkanContinuousCaptureSession.Proxy](#)>.OnSessionRequestSet ,

[CaptureSessionBase](#)<[VulkanContinuousCaptureSession.Proxy](#)>.OnSessionRequestSetWithId ,

[CaptureSessionBase](#)<[VulkanContinuousCaptureSession.Proxy](#)>.OnSessionRequestFailed ,

[CaptureSessionBase](#)<[VulkanContinuousCaptureSession.Proxy](#)>.OnSessionClosed ,

[CaptureSessionBase](#)<[VulkanContinuousCaptureSession.Proxy](#)>.NativeProxy ,

[CaptureSessionBase](#)<[VulkanContinuousCaptureSession.Proxy](#)>.TryAbortCaptures(out

[CaptureSessionBase](#)<[VulkanContinuousCaptureSession.Proxy](#)>.ErrorCode) ,

[StatefulResource.State](#) , [StatefulResource.WaitForInitialization\(\)](#) ,

[StatefulResource.WaitForInitialization\(TimeSpan, Action, WaitTimeoutMode\)](#) ,

[StatefulResource.WaitForInitializationAsync\(CancellationToken\)](#).

Extension Methods

[JNIExtensions.ToJava\(object, Type\)](#) , [JNIExtensions.ToJava<T>\(T\)](#).

Remarks

Texture conversion is done through a native Vulkan plugin.

Constructors

VulkanOnDemandCaptureSession(Resolution, GraphicsFormat)

```
public VulkanOnDemandCaptureSession(Resolution resolution, GraphicsFormat textureFormat = null)
```

Parameters

resolution Resolution

textureFormat GraphicsFormat

If not specified, uses equivalent of RenderTextureFormat.ARGB32.

Exceptions

NotSupportedException

Thrown if this constructor is invoked in an environment where Vulkan is unavailable or if the current runtime's Android API Level is lower than 33 (Android 13).

Methods

OnFrameReadyNative(nint, int, long, long)

```
protected override void OnFrameReadyNative(nint acquiredBufferPtr, int bufferDataSpace, long bufferId, long timestampNs)
```

Parameters

acquiredBufferPtr nint

bufferDataSpace int

bufferId long

timestampNs long

ProcessSingleFrameAsync(CancellationToken)

Requests and processes a single frame and returns the result.

```
public ValueTask<(long, RenderTexture)> ProcessSingleFrameAsync(CancellationTokentoken = default)
```

Parameters

token CancellationToken

Returns

ValueTask<(long, RenderTexture)>

Capture timestamp and updated texture.

Remarks

The image when returned has not *actually* finished processing, but all GPU commands to do so have been enqueued.

Exceptions

ObjectDisposedException

RequestFrame()

Requests that the next frame be processed (converted).

```
public void RequestFrame()
```

Exceptions

ObjectDisposedException